

Review

Comprehensive Review and Future Research Directions on ICT Standardisation

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Abstract: Standardisation has become imperative to retaining order and development in modern society. The simplest actions, such as train timings and the width of the railroad, would be very difficult to achieve without Standardisation. Standardisation also solves problems, such as the use of mobile devices, which requires travel abroad when out of range. We perform a large-scale quantitative analysis for papers dealing with (1) standards and (2) Information and communications technology (ICT) data in three important databases, namely Web of Science, IEEE Explore, and ACM digital library, in this paper. These three databases presented 216 articles that were divided into five categories: standard-related review and survey studies, information management across hardware and software standards, energy management standards, machine learning model classification performance, privacy-aware software system standards, and health information and communications technology standards. This paper discusses how Standardisation facilitates the planning of the entire research and innovation process by encouraging discussions regarding the specific outputs the research aims to achieve. The paper further illustrates that references related to standardisation within the call topics act as a crucial motivating factor in the decision to adopt standardisation. In conclusion, our contribution provides a better understanding of standards in peer-reviewed publications and an essential foundation for future research. In addition, we demonstrate that standards play an important role in innovation.

Keywords: information and communication technologies (ICT); De-facto; systematic literature review (SLR); standards; standardisation; standards institutes bodies



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1. Introduction

Standards play an important role in innovation by codifying information on the state of the art of a particular technology; they enable dissemination of knowledge, interoperability between new products and services and provide a platform for further innovation.

Standards in this ICT have become increasingly significant as the use and relevance of computers and computer networks have expanded. Individual and corporate purchasers of information processing equipment and software have begun to demand that the services and equipment they buy adhere to specific standards [1].

Standardisation is widely recognized as a crucial factor in facilitating the execution of research and innovation endeavors. These existing standards are used by hundreds of projects, allowing researchers to access the most up-to-date knowledge to help them implement the latest regulations and developments about industry practices, policies, and technologies [2].

In recent years, information and communication technology has received considerable attention. During this time period, it was determined that ICT could assist Standardisation in achieving its strategic objectives [3].

Thus, since the end of the 1990s, the pace of Standardisation processes within the information technology area has increased sharply, influencing the development of more standards than all other Standardisation activities combined [1].

Standards can have an underpinning or connecting function at the level of basic research, such as agreeing nomenclature or basic testing. Standards can help to advance research and innovation by enabling and supporting areas including design, testing, compatibility, and quality [3].

ICT standards have progressively gained prominence as a critical element of the information society, despite their humble beginnings. Standards are now widely recognized as essential instruments for the transfer of technology and knowledge, promote innovation, and are essential for the economy that they help innovation, and they are necessary tools for transferring technology and knowledge [4–6].

Standards can have an underpinning or connecting function, such as agreeing terminology or basic testing. As research and innovation move further towards development, so can standards by enabling and supporting aspects such as design, testing, compatibility, and quality [7].

Technology standardisation is the process of positioning applications and IT infrastructure to a baseline of standards that fit the business strategy, security policies, and goals. Standardized technology reduces complexity and offers benefits such as cost savings through economies of scale, ease of integration, improved efficiency, and better IT support. It also simplifies management of IT [8].

Standards may encompass national, European, and international levels. Numerous standards initially originate at the national level to cater to local requirements but subsequently gain broader acceptance across Europe through the involvement of organisation such as CEN, CENELEC, and ETSI. Further, these may also be the subject of international standards under ISO, IEC, and ITU-T, thus leading to the development of international knowledge sharing among nations and the elimination of technical barriers in international trade [9,10]. The national standards in the countries of CEN and CENELEC members are based on European standards. As a fundamental element of the European single market, they exert significant influence on the European economy.

Standards are said to be of a voluntary nature since they represent the associated work of many professionals from business, industry, academia, and society. These experts engage in a consensus-building and consultation process with a Standardisation organization to establish these agreements [11]. Standards serve as the fundamental framework for trade and have the potential to foster innovation by facilitating the sharing of knowledge, promoting optimal methods, and ensuring compatibility. They symbolize the principle of equal opportunity, enabling users of varying sizes to participate in fair competition. Voluntary standards possess the capacity to complement regulatory measures.

The primary objectives of this study are as follows: (1) to investigate the conventional and recent developments of relevant state-of-the-art Standardisation of research; (2) to understand the role of Standardisation; (3) to understand the need for Standardisation to improve quality in various domains; (4) to understand the role of Standardisation requirements and challenges of the ICT sector; and (5) to identify the role of Standardisation of technology and its impact on its environment.

A number of specific study topics further defined the objective. These research topics are elaborated below and were used as a basis for the in-depth research and analysis tasks performed, as described in the section.

The novelty and contribution of this paper form the crucial basis for further research by providing an overview of the place of standards within peer-reviewed publications. It covers all important concepts of Standardisation, like Standardisation bodies, standards catalogue, and other relevant standards, with detailed examples. This work also investigates the compatibility of hardware and software from different vendors and the standards regulating them. The paper also discusses in great detail the increasing emphasis being placed on the creation of standards for software systems created out of the need to address growing concerns about software privacy.

The paper also addresses the purpose of health ICT Standardisation and its role in facilitating interoperability and compatibility across diverse systems. Ultimately, the findings

highlight three key components in the literature: the obstacles to effectively utilizing these applications, the recommended measures to address these challenges, and the motivations behind the proposed general framework for the search and browsing process.

The standards principles are explained and analyzed in Section 2 of this article. The remainder of this article is organized as follows. The methodology, comprising the information source, standards for research eligibility, the systematic literature review (SLR), and the implications of publications' search results are all defined in reference Section 3. Validity risks trigger significant obstacles to the effectiveness of SLRs besides identifying research questions. In this context, the research questions are organized into four groups concerning any object from their online papers: literary taxonomy, Standardisation technology, and ICT. Section 4 is described and summarized with survey and review papers based on the most recent state of the art in Standardisation and ICT areas. We present a fresh research prospect and opening for the Standardisation area together with ICT in Sections 5–8; Section 9 discusses the motivation, challenges, suggestions, and a contemporary strategy for the Standardisation area together with ICT field, where the conclusion is presented in Section 10.

2. Preliminary Study

This section presents the fundamental concepts of standardisation, institutes bodies, standards catalogues, and other standards.

2.1. What Is Standard?

Standardisation establishes and enforces regulations and specifications to ensure conformity, uniformity, and compatibility across a particular field or industry

There can be no unique definition of standards and Standardisation adopted by all, as efforts are underway to establish a huge number of different types of standards at national, regional, and international levels in industries like agriculture, transportation, nanotechnology, medical devices, procedures, products, services, and methods. That's why it's impossible to settle on a single definition.

2.2. Standards Institutes Bodies

Standards organisation often focus their efforts on specific industries characterized by complex and specific requirements while also establishing and disseminating guidelines for diverse disciplines. There are several well-known standards organisation that can be identified, namely the following:

National Institute of Standards and Technology (NIST): is a US government department that contains a massive set of software development and security policies, guidelines, and norms that are constantly updated and publicized.

The International Electrotechnical Commission (IEC): is a global standards body responsible for developing and publishing international standards for all electrical, electronic, and related technology.

International organization for Standardisation (ISO): is a non-governmental international entity that boasts a membership of 168 national standards bodies. The organization is dedicated to the development and publication of international standards.

IEEE Standards Association (IEEE SA): is widely recognized on a global scale for its establishment of standards. The organization facilitates the development of consensus standards by employing an inclusive approach that involves active participation from industry representatives and a wide range of stakeholders.

ISO/IEC: are organisation that develop, promote, and maintain industry standards. The focus falls on communications, information technologies, and electric and electronics. Some documents are free of charge to download.

2.3. What Is Standard Catalogues?

We have reviewed the existing standards from multiple standard associations (IEEE AS, ISO, IEC, ISO/IEC). We designed a protocol and followed it to read the standards.

Standard catalogues are compilations of publications that contain standards linked to a given industry, field, or topic. This catalogue can be a valuable resource for individuals, organisation, and scholars looking to obtain and use important standards.

Example: Structure of standards institutes bodies (standards catalogues) To understand and interpret a standard, the following steps can be adhered to, as shown in Figure 1:

- A standard is reviewed every 5 years;
- Technical committees are organised through working groups, which undertake the standards development;
- Identify the standard: determine the specific standard you want to read. Standards are typically identified by an alphanumeric code or a name that reflects the subject matter or industry they cover;
- For example, ISO 9001 is a standard for quality management systems, and IEEE 802.11 is a standard for wireless local area networks (Wi-Fi).

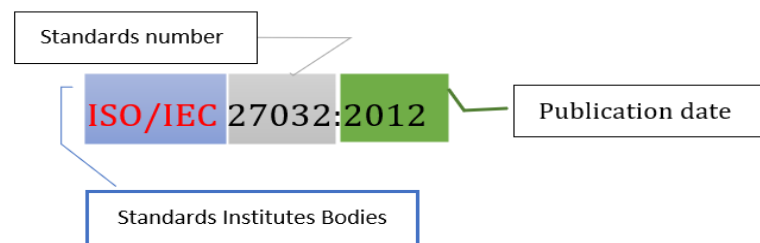


Figure 1. Structure of standards institutes bodies (standards catalogues).

Table 1 comprises the current catalogue standards, standard title, standard status descriptor, indicators, and statements.

Table 1. Overview of current catalogue standards.

Catalogues	Standard Title	Status Descriptor	Indicators	Statements
Standards Institutes Bodies. Ex: ISO/IEC xxxx	The standard title is a concise key that describes the key of name, topic, sub-topic, and type or direction.	A brief description of the status of the standard like 1. Technical committee? 2. Status of the catalogue (revised, withdrawn, published, active, deleted), 3. Publication date, 4. Link to the standard website.	The standard family (the standard comes from inheritances)–Life cycle and new publication data.	Describe in outcome terms the critical areas of competence covered by the standard.
ISO/IEC 8652	Distributed systems Annex, Information technology, programming languages, Ada.	ISO/IEC JTC 1/SC 22: withdrawn; revised by ISO/IEC 8652:2012	Life cycle and new publication data: Yes	The DSA was created to reduce the number of source changes required to transform an ordinary non-distributed program into a distributed application.
ISO/IEC 27032	International Electrotechnical Commission, Information security, cybersecurity, and privacy protection.	ISO/IEC JTC 1/SC 27: published	Life cycle and new publication data: Yes	Provides guidance for improving the state of cybersecurity, drawing out the unique aspects of that activity and its dependencies on other security domains.
ISO/IEC 2382-1	Information technology (IT) in general information technology vocabulary; Part 1: fundamental terms.	ISO/IEC JTC 1: withdrawn; revised by ISO/IEC 2382:2015	Life cycle and new publication data: Yes	Information representation, hardware, software, programming, applications and end users, computer security, and data management are all examples of general terminology.
ISO/IEEE 11073-10101	Part 10101: point-of-care medical device communication, nomenclature in health informatics.	ISO/TC 215: published; revised by SO/IEC 2382:2015	Life cycle: Yes	The primary focus is acute care medical equipment and patient vital signs data.
IEEE 802.15.4	Information technology, wireless networks.	C/LAN/MAN-LAN/MAN: published	-	Physical layer (PHY) and medium access control (MAC) sublayer specifications enable low-data-rate wireless connection with fixed, portable, and moving devices with minimal battery consumption.
ISO 19650-1	Building and civil engineering information organization and digitization, including (BIM) information management utilizing building information modeling.	ISO/TC59/SC13: published	-	Life cycle of built asset includes planning, design, engineering, construction, operation, maintenance, refurbishment, repair.

2.4. Other Standards

A “de facto standard”, also known as a “standard in actuality”, occurs when a successful solution is widely and independently adopted by multiple industries within a market segment, and products made on that basis are universally accepted by customers. Many technologies, products, and services are based on existing standards, therefore formal and de facto standards have an impact on our daily lives.

Example 1: De facto standards have the potential to transition into formal standards when they are officially published by recognized standards development organisation (SDOs). Examples include HTML and HyperText Markup Language, which were developed by Tim Berners-Lee at CERN in Geneva, Switzerland, during the beginning of the 1990s and have since been under constant maintenance by the W3C.

Example 2: Another example is PDF (Portable Document Format), which was created by Adobe Systems in 1993 and subsequently standardized by ISO (ISO 32000, ISO 19005-1:2005).

Example 3: QWERTY, the most used keyboard layout today, was first patented in 1864 by Christopher Sholes. Later, in 1936, August Dvorak came up with its successor, called Dvorak, intended to increase typing speed. The QWERTY had been so successful that people just didn’t switch to this new design, even though most modern operating systems support Dvorak natively.

2.5. Research Questions

Research questions are essential in a systematic literature review. In particular, the determination of the search strategy and analysis for this study was based on the following research questions (RQs):

RQ1. What does the current research literature reveal about standardisation and ICT methods and topics?

RQ2. What are the main goals, vision, and trends in Standardisation, and which research topics are most prominent in this field?

RQ3. What are the existing research gaps in the field of ICT Standardisation?

RQ4. What are the key motivations, concerns, challenges, and recommendations for improving Standardisation and ICT methodologies?

RQ5. How can standards play a more prominent role in achieving regulatory objectives and fostering innovation?

3. Materials and Methods

As stated in the introduction, we adopted an SLR to gather the research articles available that supported two concepts: “standard” and “information and communications technology”. The research articles were collected through three well-known digital libraries as follows: (1) Web of Science—which provides multidisciplinary research articles on scientific topics; (2) IEEE Xplore—which deals explicitly with research articles on electrical and electronics engineering; and (3) the ACM digital library, which provides a comprehensive database for computing and information technology.

In March 2023, many searches on the three databases were conducted using keywords (or phrases) such as “standard”, “standards”, “de facto standard”, “Standardisation”, “information and communications technology”, “ICT”, “software”, “application development”, “Apps”, “techniques”, “methods”, “implementation”, “guide”, and “algorithm”. The keywords differed relatively slightly. Following that, conjunctions “OR” and “AND” were used to link these terms, which were then followed by “refining information”.

Figure 2 depicts the search queries utilized in this experiment. Some results were removed because they were letters, magazine pieces, or book chapters. The primary purpose of this exclusion was to gather the most up-to-date scholarly literature on the standard difficulties and opportunities. The results were then separated into two categories: (1) general and (2) coarse-gained. The paper is discussed in five succeeding sections derived from the study findings, where Google Scholar SE was used to establish the study's direction.

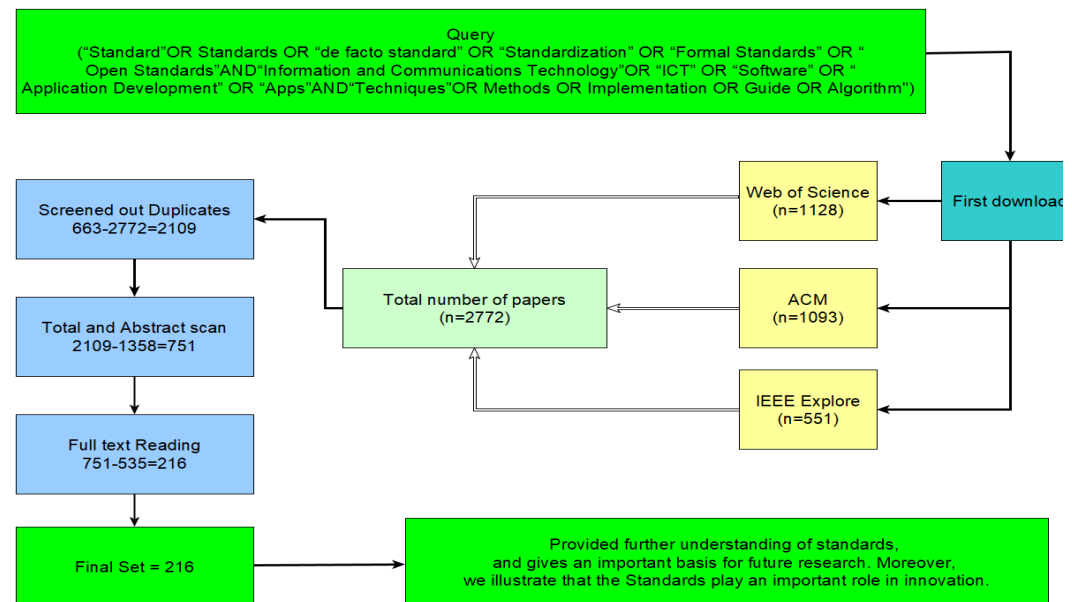


Figure 2. Research methodology guideline.

The importance of the collected articles was assessed in order to keep the most relevant articles from a huge number of collected academic papers. Furthermore, they included articles were classified using two criteria: (1) initial screening to find relevant results and (2) three iterations in the filtering process to remove redundant and duplicated articles.

As previously stated, an article was excluded if it did not meet the following selection criteria: (1) the English language was not used to write the paper; (2) standard and/or Standardisation were the focus of the paper; and (3) the research interest in the paper was only focused on standard without information and communications technology. Furthermore, after the second exclusion cycle, the articles could be removed if (1) the contribution of the paper did not consider any aspects of the standard on information and communications technology, or (2) the discussion in the paper was solely focused on standard on information and communications technology and did not consider any other topic.

Statistical Information of Collects Articles

The outcome of the review is addressed in the form of answers to the research questions.

RQ1. What does the current research literature reveal about Standardisation and ICT methods and topic?

In this effort, papers were filtered extensively before being classified into five groups based on the proposed methodologies and future research plans on Standardisation. The categories were the following: (1) review and surveys, (2) computer communication networks standard, (3) smart technology standard, (4) security techniques standard, and (5) health informatics standard. Subsequently, further subcategorization was performed according to the authors' writing and presentation of the articles to readers.

Our findings are shown in Figure 2, where we found 2772 research papers based on user searches, of which 1128 were found in WOS, 551 in IEEE, and 1093 in ACM digital libraries. The chosen articles were all released between 2013 and 2023. Then, these articles were split into three groups: (1) the 663 redundant articles; (2) the 1358 irrelevant articles based on the titles and abstracts; and (3) the 216 articles that met the standards for ICT search criteria.

The statistics for the various categories listed above for the articles about ICT Standardisation are shown in Figure 3. In the figure, it can be seen that the 216 articles from the three databases were divided into review and surveys (27), computer communication networks standard (126), smart technology standard (19), security techniques standard (19), and health informatics standard (25).

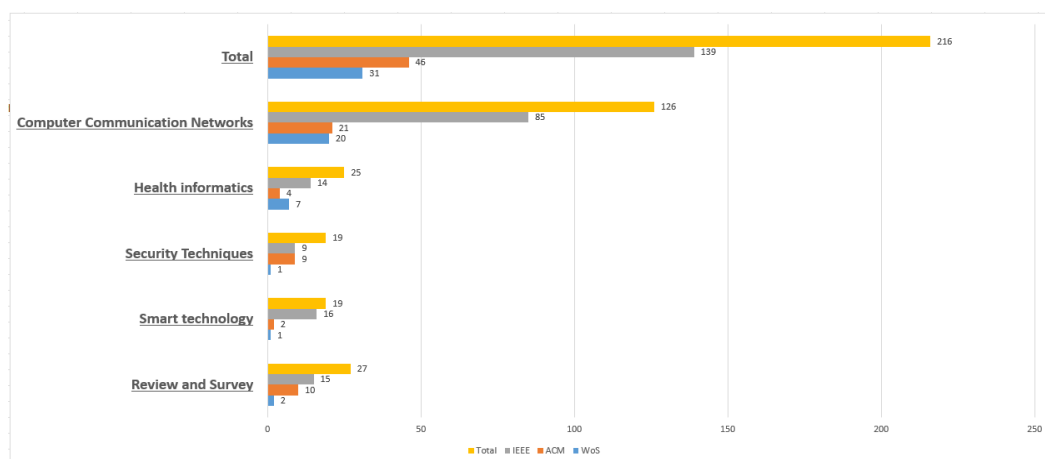


Figure 3. Articles categorized based on their contribution.

Figure 4 depicts the statistics for the publications released between 2013 and 2023. The figure depicts the number of research articles published in each of the five categories for each year. It can be noted that forty-four papers were published in the early years, such as in 2013. In both 2014 and 2015, the value remained consistent at 23. However, in 2016, there was a significant drop to 9. This decline was followed by a sharp increase in 2017, where the value rose to 25. The upward trend continued in 2018, reaching 27, and declined again to 22 in 2019. From 2020 onwards, the number of publications remained stable until 2023, when the number of publications reached 32 research articles. This demonstrates a growing study tendency toward the Standardisation notion.

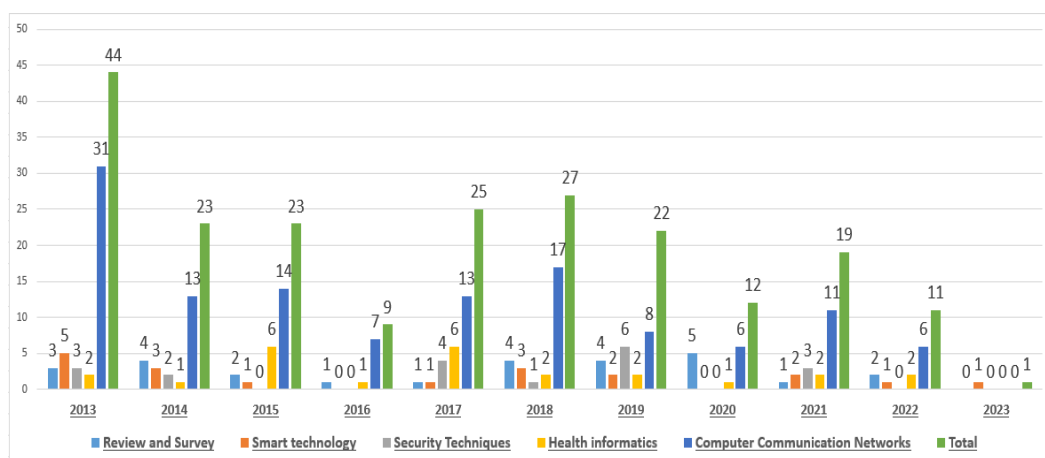


Figure 4. Statistics of publications released between 2013 and 2023, including year-by-year data on publication frequency.

Regarding distribution by author geographical coverage, Figure 5 illustrates the implementation of Standardisation by the utilization of standards in 56 countries and likewise provides the authors' nationalities. We observed that the relevant studies were conducted geographically, where attempts were made to adopt standards to cover cases in research. The distribution of the 216 articles in standards shows that the most productive authors are from the following countries: USA and China (21); Italy (15); Germany (12); Korea and Spain (11 each); India (10); Brasil, Sweden, Japan, and the UK (8 each); Pakistan, Russia (5 each); Finland, Austria, Australia, Canada (4 each); Portugal, Thailand, Belgium, Greece, New Zealand, Turkey, Netherlands, Ireland, France, Malaysia (3 each); Switzerland, Croatia, Singapore, Taiwan, Hungary, Iran, Romania, Ecuador, (2 each); Bangladesh, Bosnia, and Herzegovina, Hong Kong, Norway, Saudi Arabia, Sudan, Indonesia, Senegal, Costa Rica, Poland, Bulgaria (1 each).

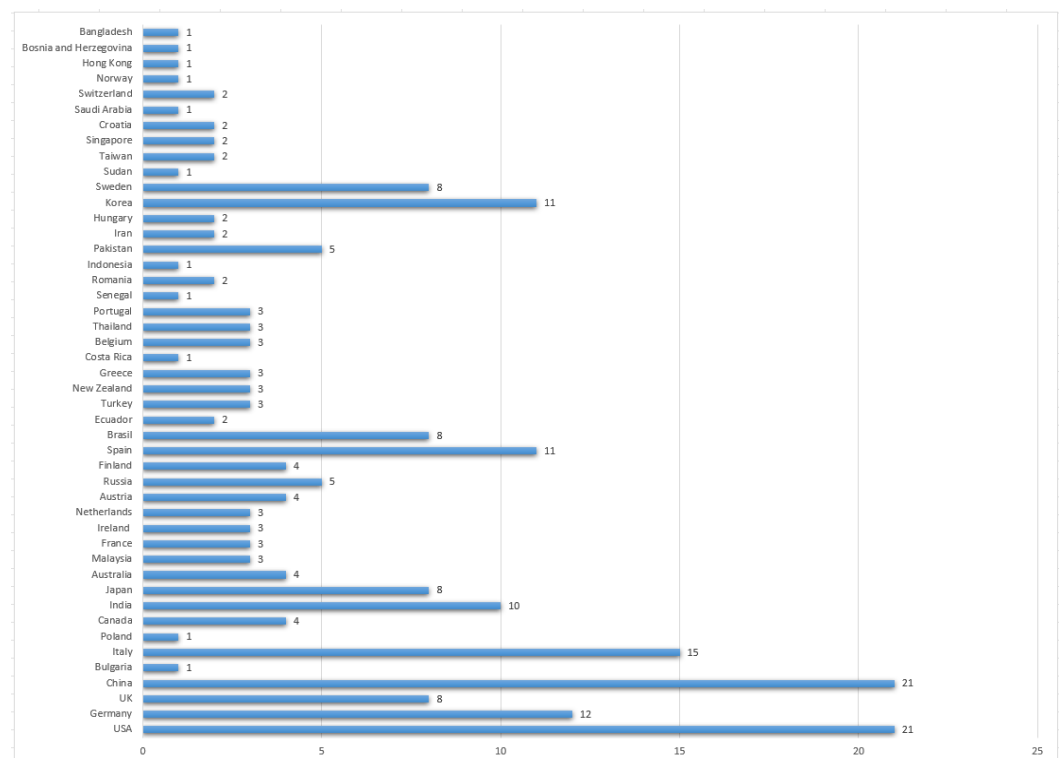


Figure 5. Distribution of publications by authors' geographical coverage from 2013 to 2023.

We created a Standardisation taxonomy representation of the current literature in light of comprehensive Standardisation produced in recent years, as shown in Figure 6. The presented approach consisted of several Standardisations, including systems and software engineering standard, computer-communication networks standard, smart technology standard, security techniques standard, and health informatics standard. explaining the proposed system in detail with the help of a block diagram.

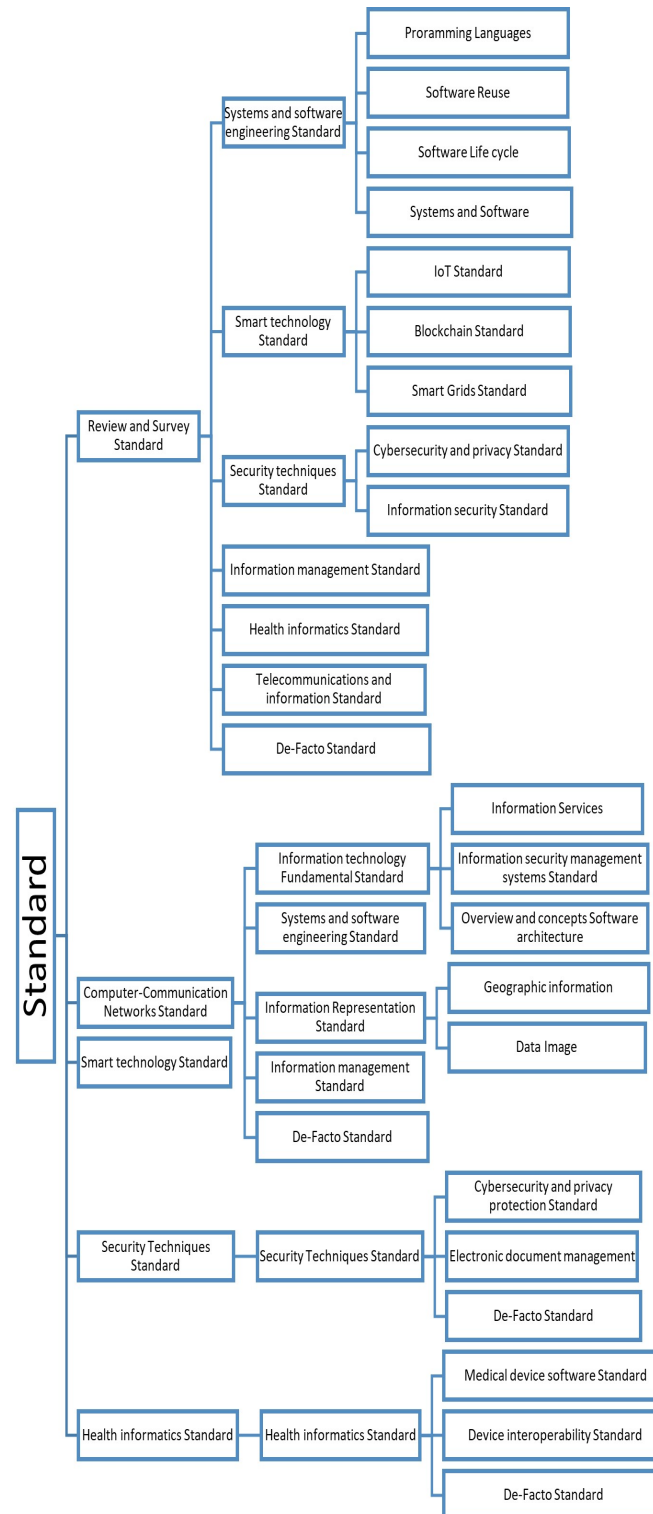


Figure 6. axonomy of literature on Standardisation.

4. Review and Survey Articles

Recently published survey and review articles provide a comprehensive description and summary of the current state of the art in Standardisation and its applications in the area of education research. The research reviewed the technological hurdles and Standardisation considerations. This category's twenty-seven articles were broken into seven subcategories.

RQ2. What are the main goals, vision, and trends in Standardisation, and which research topics are most prominent in this field?

The topics can be categorized as follows: computer-communication networks standard, smart technology standard, security techniques standard, and healthcare informatics standard.

4.1. *Systems and Software Engineering Standard*

The analysis and research documentation provides an overview of the current understanding of Standardisation in systems and software engineering technology readiness

4.1.1. Programming Languages Standard

A programming language standard is a formal document that specifies the syntax and semantics of a particular programming language. Typically, when discussing real-world languages, this document entails a detailed explanation using easily understandable language rather than formal semantics expressed in mathematical terms.

Two articles [12,13] were presented that conducted a survey on real-time distribution middleware, focusing on its adherence to standards and its impact on the industry. The objective of these articles was to offer technicians and domain experts a comprehensive understanding of this field while also highlighting important unresolved matters and suggesting potential avenues for future research. It also discusses the preliminary measures implemented in the utilization of distribution standards in high-integrity systems. The survey concludes by providing a comprehensive analysis of the advantages and disadvantages associated with these various approaches. It also highlights any unresolved matters, significant challenges, and potential avenues for future research.

Other articles [14] outline how hygienic and partially hygienic macro technology became adopted and enormously impacted the scheme. The interplay between Standardisation push and algorithmic progress forms a particularly salient part of the narrative.

4.1.2. Software Reuse Standard

Another subject in this section, Standardisation of software reuse, results in increased reliability, productivity, quality, and cost savings. Current reuse strategies focus on the reuse of software artifacts, which is based on projected functionality, but the non-functional (quality) side is also significant.

An article [15] was published that provides an expert review of a framework developed for the implementation of a novel testing approach known as taxonomy-based testing (TBT) in the domain of medical device software. This framework presents three methodologies for implementing TBT and has undergone validation by experts from the software testing industry and the medical device software domain.

A comprehensive analysis was conducted on multiple software engineering standards issued by the institute of electrical and electronics engineers standard association [16]. An investigation was conducted on software engineering standards to identify the factors and sub-factors that may contribute to diverse situations among members of software development teams. Consequently, a total of 12 factors, along with 52 sub-factors, have been identified as potential contributors to diverse situations among members of software development teams

4.1.3. Software Life Cycle Processes Standard

The implementation of standardized system lifecycle processes as a means of enhancing efficiency is centered on attaining a consistent level of delivery at the enterprise level throughout every stage of the lifecycle. Furthermore, a standardized approach to system lifecycle management not only offers a structure for effectively overseeing individual systems-related tasks, but also enables the organization to gain comprehensive insight into the utilization of system-related resources, financial aspects, and project progress at an organizational level.

Two articles [17,18] present research conducted on the topics of governance, management, conceptualization, evaluation, elaboration, and enablement. The set consists of 45 activities, each containing a specified number of tasks ranging from five to eight. This

standard can serve as a fundamental basis for organizational training and guidebooks pertaining to architecture management and development. A preliminary version of the model's architecture and its components is constructed, yielding partial results. The model is developed based on key references and best practices identified for agile governance during the course of this research.

The author of study [19] employs a SLR methodology to identify and analyze the strategic process improvement (SPI) initiatives within small and medium-sized enterprise (SME) software firms. The findings indicate that a significant proportion of software process improvement initiatives within small and medium-sized enterprises (SMEs) are implemented in the regions of America and Europe. The SPI initiative was exclusively considered in Hong Kong, a country located in Asia, where the CMM Fast-Track model was developed. Numerous proposals have been put forth regarding software process improvement (SPI) initiatives for small and medium-sized enterprise (SME) software firms.

4.1.4. Systems and Software Quality Requirements and Evaluation (SQuaRE) Standard

The topics covered in this document include general guidelines, planning and management, system and software quality models, data quality models, quality measurement frameworks, quality measure elements, framework evaluation processes, evaluation guides, evaluation modules, testing, and common industry formats.

An article [20] proposes a software evaluation technique based on user feedback and the data existing in the help desk database system. It requires processing the questions for assistance sought from the help desk over time to identify the root cause of issues and recommend product and methodology improvements to reduce the occurrence of similar questions in the future. The analysis is conducted using the ISO9126 quality model and incorporates the utilization of ISO 31010 tools to identify the underlying causes of system issues within the production environment.

4.2. Smart Technology Standard

The term “smart” pertains to the various formats, processes, and tools that are essential for facilitating user interaction with standards. This interaction can involve both human- and technology-based users. These digital solutions cater to the requirements of various stakeholders, including industry players, regulators, end users, and society.

4.2.1. Internet of Things (IoT) Standard

The lack of standards in the IoT makes the interoperability problem between different systems and devices work well together. It is tough to fit just one standard onto all IoT devices since they can use various communication protocols and data formats, and many vendors may make them.

Article [21] focuses on strategies to improve digitization and Standardisation within the context of Industry 4.0. This review examines both traditional and recent advancements in IIoT technologies, frameworks, and solutions that aim to enhance interoperability among various components within the IIoT ecosystem. The work also engages in discussions regarding various interoperable IIoT standards, protocols, and models that can be utilized to digitize the industrial revolution.

4.2.2. Blockchain Standard

Blockchain is a decentralized, immutable database for tracking assets and transactions in corporate networks. It enables the recording and trading of value, reducing risk and increasing efficiency.

Article [22] was presented that offers a comprehensive review of grid and prosumer blockchain applications while also highlighting the standard development activities conducted by the IEEE SA. The review indicates that blockchain technology in the energy sector is in a state of maturation and holds potential for application across various domains

within the power and energy industry. It has the capacity to facilitate the advancement of diverse modern grid technologies.

4.2.3. Smart Grids Standard

A smart grid is an electricity network that monitors and manages the conveyance of power from all generation sources in order to satisfy the shifting electricity demands of end users. The number of successful smart grid initiatives continues to expand, with each project ranging in scale and complexity.

Article [23] on using IEC 61850 with the IEC 61499 references a model for distributed automation and creating a smart grid compliance profile. The IEC 61850 power utility automation standard ensures strong interoperability, which smart grid and its devices and parts need.

4.3. Security Techniques Standard

The goals of Standardisation are uniformity, tracability, and repeatability. By always using the same security methods across all work, security knows what is protected and what is not. This lets them know if they need to take extra steps and lets them know if there are any exceptions. Standardizing security policies can take many forms, such as legal compliance, access controls, acceptable use policies, security as code, and automation, to name a few.

4.3.1. Cybersecurity and Privacy Protection Standard

A cybersecurity standard refers to a collection of guidelines or best practices that organisation can adopt in order to enhance their cybersecurity stance. organisation have the opportunity to leverage cybersecurity standards in order to effectively identify and implement suitable measures for safeguarding their systems and data against potential cyber threats. Standards can also offer valuable guidance regarding the appropriate actions to take in response to and mitigate the impact of cybersecurity incidents.

One article [24] described the outcomes of implementing different lattice-based authentication algorithms on smart cards and a microcontroller often found in smart cards. Only a handful of the proposed lattice-based authentication protocols may be implemented utilizing the restricted resources of such small devices; however, cutting-edge ones are sufficiently efficient to be employed practically on smart cards.

Paper [25] presented a comprehensive review of the key attributes of wireless communication technologies in the context of Industry 4.0. Additionally, it examined various security vulnerabilities and concerns that exist within this sector. Additionally, a concise summary was provided for each of the industrial revolutions, along with an emphasis on the primary technological distinctions that set them apart.

4.3.2. Information Security Management Systems Standard

Data security standards are a set of guidelines or criteria to which an organization can adhere in its attempt to secure sensitive and confidential information against unwarranted access, use, disclosure, disruption, modification, or destruction.

The aim of study [26] is evaluation and analysis of different methods and standards in secure software development. This can be achieved by surveying and comparing to identify at which specific stages of SDLC one or more of these methods/standards can be used to make a software application more secure during its development process.

4.4. Information Management Standard

The purpose of the information management standard is to establish fundamental principles and obligations that must be adhered to in order to achieve efficient information management. The purpose of this system is to ensure that information is created, managed, retained, and disposed of in a manner that aligns with appropriate protocols and guidelines.

Article [27] discusses the capabilities and requirements necessary to ensure information quality in the context of reducing vulnerabilities in DAECO. This study incorporates the theoretical frameworks of resilience, information quality, and vulnerability, as well as the established standards of ISO19650 and ISO27001.

An additional study [28] was conducted, which presents a SLR consisting of 47 carefully chosen user studies for the field of end-user software engineering (EUSC). The article employs a review framework to systematically and consistently evaluate the focus, methodology, and cohesion of each of the aforementioned studies. The results contribute to the creation of a design framework and a series of inquiries for the purpose of designing, reporting, and evaluating effective user studies for EUSC.

Paper [29] introduces and summarizes work in the IVHM/PHM standard structures and standards. Many standard organisation in the IVHM/PHM domain are eager to build their standard systems for various purposes and applications, and they form various standard structures. The most important organisation are ISO, SAE, ARINC, and IEEE.

4.5. Health Informatics Standard

Health informatics assist in ensuring the safety, usability, and appropriateness of health-specific applications utilized by clinicians, patients, and citizens. Interoperability is advocated for the purpose of facilitating accurate interpretation and utilization of electronically communicated information across systems, thereby enabling effective decision-making, continuity of care, and other relevant objectives.

According to one publication [30], support for graphical-based assurance case assessment is included in 10 assurance case tools, all of which have remarkable assessment capabilities. More specifically, by studying the evaluation techniques supported by the tools, three kinds of methods for reviewing an assurance case's structure and five categories of assistance for assessing an assurance case's content were discovered.

The purpose of study [31] is to conduct a SLR that examines the level of support provided by current standards, specifically ISO 13485:2016, ISO 14971:2012, IEC 62304:2006, and IEC 62366:2015, as well as current reference architectures for medical device software, towards ISO/IEC 25010:2011 and the 4+1 views.

Article [32] aimed to discuss the issues of ensuring the safety of product lines and detecting possible insufficiencies in already available research studies. An SLR of research publications published up until January 2016 was conducted. The task was to identify 39 research articles to provide a complete list of primary studies. The objective was to conduct an analysis of how product lines are documented within these articles, with a particular focus on the safety-related topics that are covered and the evaluation methods employed by the studies.

Two articles [33,34] provided a comprehensive analysis of currently employed software-based tolerance techniques that effectively safeguard the system against soft errors. The aforementioned techniques were subsequently linked to the software requirements mandated by EN 50128, which serves as the software standard within the railway industry, as well as those mandated by IEC 60601-1, the overarching standard for the medical field. The detection techniques were demonstrated to facilitate compliance for both EN 50128 and IEC 60601-1.

4.6. Telecommunications and Information Standard

This converged communication marketplace is rapidly combining voice, data, video, and wireless technologies. The more an organization becomes a part of this vibrant communications network, the more it realizes quickly that Standardisation, planning, and continuous improvement are veritable requirements for functionality, interoperability, and reliability.

Study [35] was conducted using mmWave communication and radar devices in in-device-based localization and device-free sensing with a view to indoor deployments. The paper provided insight into fundamental concepts relevant to mmWave signal prop-

agation and system design. It intended to cover a number of approaches, algorithms, and applications for mmWave localization and sensing.

Another article [36] surveyed the importance of old and new TMN and ICT management. Additionally, it is necessary to identify all the prevalent networks and technology management systems known today to create a basis for designing network management architectures.

4.7. De Facto Standards

In one article [37], the state of computing is surveyed in relation to 3D virtual spaces, and the steps necessary to create a network of 3D virtual worlds, or metaverse, that provides a compelling alternative realm for human sociocultural interaction are outlined. Convergence on the network protocol used by Linden Lab for Second Life as a de facto standard has enabled decentralized development, which in turn has led to the decoupling of the client and server sides of a virtual world system.

Article [38] presents a comprehensive survey that focuses specifically on algorithmic and combinatorial aspects. The majority of routing protocols ensure the achievement of convergence towards a stable state of routing. In the event that there are no alterations to the topology or configuration, it can be observed that each router ultimately establishes a consistent route to any given destination. However, this assertion does not hold true for policy-based routing protocols, such as the Border Gateway Protocol (BGP), which is widely accepted as the standard for interdomain routing.

5. Computer Communication Networks Standard

The second category included 126 papers related to the objective to ensure compatibility between hardware and software from various vendors. Progress in the development of information-sharing networks would have been very hard, if not impossible, without networking standards being established. Customers can also have the flexibility to choose their desired vendor from the various vendors available on the market.

Customers have the flexibility to procure hardware and software from any vendor, as long as the equipment adheres to the established standard. Standards play a crucial role in fostering increased competition and maintaining price stability.

5.1. Information Technology Fundamental Standard

The scale of information technology standards relates to wide-ranging hardware and software standards. The importance of software and information formatting standards is steadily increasing. There are defined standards for operating systems, programming languages, communications protocols, and human–computer interaction. As one simple example, it quickly became apparent that standard protocols are needed if electronic mail messages are interchanged worldwide. These protocols define addressing schemes, message formats, and efficient transmission techniques.

5.1.1. Information Services Standard

This section outlines the necessary requirements that a service must meet in order to demonstrate its suitability for the intended purpose. The standard may encompass various components, including definitions, indicators of service quality, and their corresponding levels, as well as stipulations regarding the timeframe for service delivery. An example of such a standard is one pertaining to the management of customer complaints.

A Warbler family, a pseudorandom number generator family based on NLFSRs with acceptable randomness, is presented in [39]. The security analysis demonstrates that the proposed instances can pass the cryptographic statistical tests recommended by the EPC C1 Gen2 standard and NIST, as well as resist cryptanalytic attacks such as algebraic attacks, cube attacks, time-memory-data tradeoff attacks, Mihaljevi et al.'s attacks, and weak internal state and fault injection attacks.

One paper [40] describes software for analyzing 3D-scanned electronic geometric models. ISO 8559-1 offers specialized information equivalent to national regulatory texts. It briefly discusses basic information completeness and presentation needs.

Another reason is the lack of thorough test data and the flexibility of these procedures. Based on extensive outdoor test data, software scripts are used to evaluate the performance of these three standard translation methods recommended under IEC 60891, where the reference conditions are optimized to produce the best result in different irradiance and temperature ranges [41].

An experiment in [42] shows how to develop a two-level warning system, cautionary and imminent, against a driver's lane change intentions. Experimental results presented there show that an in-vehicle safety warning system provides adequate precautions against a driver's lane change. This is achieved through the use of visual and audible warnings in accordance with the ISO17387:2008 standard.

Two articles [43,44] use the CPM/PDD to study UCD. Many user-centered design methodologies are based on the international standard 9241 210. It investigates the mapping of the two models in order to acquire support and insights for selecting methodologies and metrics for the creation of interactive systems. Also detailed is the creation of a PHR-S for MS management based on the concepts and recommendations of the ISO 9241-210 standard.

5.1.2. Electronic Document Management Standard

Managing information across the organization aids in the support of a variety of new technologies that produce electronic records standards, and record generation can be both active and passive (e.g., automated logging of system upgrades). Individual records standards can be established via electronic mail systems, as web-based publications, or as documents published and held in administrative information systems.

One study [45] offers findings concerning standard specifications and their implementations in open-source software. Specifically, the analysis focuses on extensive insights and experiences connected to two open-source projects that implement the PDF format standard.

Another paper [46] on the finite element method using CST simulation software to design a GTEM cell before it is put into use is presented. After the GTEM cell is put into place, a three-axis electromagnetic probe is used to test how regular it is. According to the IEC 61000-4 and IEEE 1309-2005 standards, the testing area and volume of the object being tested inside the GTEM cell are shown.

5.1.3. Overview and Concepts Software Architecture Standard

This standard outlines the specifications for exception conditions and their corresponding default handling procedures. A floating-point system that supports this standard can be implemented in either software, hardware, or some combination of both.

The proposed [47] methods for floating-point arithmetic are built on the IEEE 754-2008 standard. These methods are a mix of software and hardware. Software that makes good use of space is used to create multiplication. Based on platform analysis, the proposed approaches can reduce execution time by 24–81% compared to a typical software library and utilize 25–94% less hardware than an area-efficient full decimal multiplier.

Two publications [48,49] on the floating point arithmetic cores offered by major FPGA manufacturers are not completely compliant with IEEE 754 standards. These circuits cannot accommodate non-normal numbers. Also, these papers describe the implementation of a single precision floating point adder compliant with IEEE 754 that accepts denormal inputs. In addition, its performance and resource consumption is compared to that of the Xilinx floating-point adder IP core.

5.2. Systems and Software Engineering Standard

A software engineering standard can be characterized as a standard, protocol, or analogous document that delineates the regulations and procedures governing the develop-

ment of software products. A conventional software development company typically maintains a collection of documents that are exclusively intended for internal use within the organization.

5.2.1. Programming Languages Standard

Standards are measurable criteria that everyone agrees to use so that they can communicate with each other. Some people think that the first standards were monetary systems that were implemented to help people trade things. A language is a way to talk to each other. The alphabet is a common way for people to share knowledge.

One document [50] describes adaptations of extant approaches to dialog act recognition proposed in order to address the hierarchical classification issue. Specifically, the proposed end-to-end hierarchical network predicts communicative function, maintains dependencies, and determines stopping level using cascading outputs and maximum a posteriori path estimation.

A recent study [51] introduces a modeling language that utilizes the ISO/IEC 24744 metamodel. This language effectively represents textual information in a well-structured format, allowing for the description of semantic relationships within the discourse. Furthermore, the paper presents an application of the proposed language in the field of requirements engineering, demonstrating the advantages of implementing the suggested approach and its potential in various textual domains.

A recent study [52] on the SystemC, IEEE 1666-2005 standard for embedded system modeling, architectural exploration, performance analysis, software development, and functional verification has received SA approval for RTL simulation.

5.2.2. Software Reuse Standard

The objective was to present a set of standards for software development frameworks that enable the creation of secure and high-quality metamodels for software development methodologies. Additionally, we aimed to emphasize the alignment between each phase of software development and the relevant standards and regulations.

One publication [53] proposed techniques for discovering and extracting process models from legacy databases, laying the theoretical foundation for a model-driven framework and proposing gPROFIT, a machine learning tool for real-world application.

Another paper [54] offered an overview of TDL and discussed the ways in which it could be used in a train use case. Events can be described at a higher level of generality than with programming or scripting languages. Also, TDL can be used to describe tests that come from other places, like simulators, test case generators, or logs from previous test runs.

Two papers [55,56] proposed applying the principles of software reliability engineering to commercial software creation. They established a number of additional steps to ensure software stability at each stage of the SDLC. In addition, they utilized models and metrics for gaining access to and analyzing software product reliability. These are founded on universally accepted criteria for ensuring the safety of computer programs.

5.2.3. Software Life Cycle Processes Standard

The software development lifecycle (SDLC) is the method that development teams use to design and build high-quality software in the least amount of time and for the least amount of money. The goal of SDLC is to plan ahead to reduce project risks so that software meets customer standards during development, and afterward outlines how to set up a common framework for software life cycle processes with well-defined terms that the software industry can use as a guide.

In one study [57], a design framework called TooMVBC was proposed to turn a formal, verified computation model into an executable MVBC code. TooMVBC uses the formal computation model MVBChart to obtain a high-level idea of what the MVBC is. TooMVBC is applied to design MVBC with the highest class 5, which is what the standard IEC

61375 says to do. During formal verification of the built system model, several important ambiguities or bugs in the standard are found.

Another paper [58] described a system for modeling software development processes that ensures standard compliance from specification to execution. From specification to enactment, there is no (semi-)automatic technique to ensure de facto processes meet de jure norms.

Paper [59] presented an ISO/IEC 25040-aligned evaluation approach, activities, and artifacts for serious gaming quality evaluation. It also introduced a serious game quality assessment method (SG-QUAM). It also supplied a usability and portability model matched to the ISO/IEC 25010 standard and tailored to this area, as well as quality in use when players use the application.

Study [60] offered a viewpoint-oriented strategy for developing Industry 4.0 applications to better adapt system models to stakeholders and concerns. The technique uses existing or developing systems and software engineering (ISO/IEC/IEEE 42010:2011) and Industry 4.0 (RAMI 4.0, IEC 61131-3) standards. The study then modeled and implemented views, including an industrial use case.

The author of [61] introduced a substation data processing unit for mapping entities between power systems and communication networks using IEC 61850-based modeling.

IEEE standard 1459 uses voltage and current signals to identify power quantities, using NCA, MRMR, and CST for feature selection, reducing computational time and improving classification results [62].

Two publications [63,64] assessed software quality, an important computer software research topic according to the International organization for Standardisation for understanding, measuring, and assessing software quality. Standardisation improves software quality prediction and understanding. Introduced international quality standards development and focused on software quality development trends.

Five [65–69] small software development companies clustered in a process improvement program implemented the ISO/IEC 29110 standard. A series of papers reported the findings and lessons learned. The ISO/IEC 29110 engineering and management manuals are simple and free, which has encouraged their acceptance. Over 17 nations teach ISO/IEC 29110. VSEs may show local and international customers and partners their expertise via low-cost independent certification and assessment methods. This study showed practical frameworks for each software implementation activity, including inputs, outputs, procedures, and limitations. ISO/IEC29110 is a lightweight project management and software implementation standard.

Four [70–73] examples CERTICSys supported process assessment utilizing the ISO/IEC 15504 standard. CERTICS evaluates and certifies Brazilian software, developed and innovated. The NBR ISO / IEC 9126 standards are used to evaluate software, and the ten golden guidelines for medical education software are described. SimDeCS is used to test this method.

Some papers [74–78] used ISO/IEC standards like 9126, 14598, and 15939 to create software that integrates the theoretical elements of these standards with the practicalities of software maintenance while creating a software application. The main results were a set of uniform processes and procedures and a process asset library to support good project management practices in participating companies.

5.2.4. Systems and Software Quality Requirements and Evaluation (SQuaRE) Standard

In this section, we describe how the software product was created and how it works, in addition to its functional aspects. General guidelines, planning and management, system and software quality models, data quality models, quality measurement framework and elements, framework evaluation process, evaluation guide, evaluation module, testing, and standard industry format are all covered.

Two papers [79,80] suggested boundary scan. The IEEE1149.1 boundary scan test standard in ICs helped miniaturize PCB assembly. These papers discussed JTAG boundary

scan test solution applications and MPU JTAG provision development. They also examined how students, instructors, and professional developers regard code quality, including which features they value most.

Three studies [81–83] explored numerous quality aspects in a comprehensive model influenced by the ISO 25010, 42010, and 12207 standards for software quality, architecture, and process, respectively. They presented an ISO/IEC 25010-based quality methodology for RPG software maintainability evaluation. They also showed a case study of how a mid-size software company with over twenty years of RPG application development implemented the quality model as a continuous quality monitoring tool into its business processes.

Four papers [84–87] referenced IEEE STD 1061-1992 and ISO/IEC 25010:2011 SQuARE software quality models. Software engineering and game quality models determined game quality. Several case studies used game testing. Case studies showed how to evaluate games for relevant results. Game makers can improve games using game quality outcomes, according to studies.

Another study [88] examined the gap and proposed an object-oriented design paradigm for information security. The above-classified design patterns do not provide a path or standards to build an information system that meets confidentiality, integrity, and availability requirements.

Study [89] on the IEEE 729-1983 standard defined software quality as “the set of features of the software that determines how well it will meet the needs of the customer when it is used”. This research presented an ACO strategy that intensifies search around the metric neighborhood to increase software quality estimate model prediction accuracy. Implementation and positive results comparisons were presented.

Paper [90] anticipated E/E system functional expansion overviews COSMIC ISO 19761. Functional size can estimate development work, manage project scope modifications, benchmark productivity, and normalize quality and maintenance ratios.

An essay explored constructing a fuzzy inference rule basis to assess mobile app quality. The solution used the major provisions of the PNST277-2018 standard “Comparative tests of mobile applications for smartphones” [91].

5.2.5. Software Interfaces Standard

The user interface is an umbrella term for all of the parts of a computer system that the user can see and interact with while using the system. Information is based on standards. To describe it, both should be considered: the fixed parts, like the layout of the keyboard, panels, and reports, and the moving parts, like how programs should react to the user’s actions.

In one paper [92], the EN 50128 standard for software development for the European railway industry was used to look at agile practices for the first time. The study added to what has already been learned about using agile methods in other controlled areas.

5.3. Information Representation Standard

The comprehension of various data types and the utilization of diverse representation techniques align with the established standard and prove highly advantageous within the field of archaeology. This is particularly relevant as temporal knowledge in this domain often originates from multiple sources and is presented in various formats.

5.3.1. Geographic Information Standard

Within the field of geographic information, these standards may define data handling methods, tools, and services. People think that data management means obtaining, processing, analyzing, accessing, displaying, and posting data for users and systems.

One publication [93] based on ISO standards 19107 and 19108, which offer spatial and temporal properties for geographic features, defines a spatio-temporal model for archaeological data. In order to implement interoperability, the article proposes a spatio-

temporal conceptual model for archaeological data based on ISO standards of the 19100 family and suggests using GeoUML.

After the derivation procedure, another publication [94] provides a framework using ISO 19107 to represent and infer temporal provenance data. Fuzzy logic is used to assign a degree of confidence to values and fuzzy temporal constraint networks to model relationships between dating of different findings represented as a graph-based dataset. Archaeological data, especially temporal data, are often vague because many interpretations can coexist.

5.3.2. Data Image Standard

The establishment of a data compression algorithm involves the development of optimized data. The primary factors that impact the accuracy of this algorithm include the resolution of the image, the level of image noise, the adherence to quality and information standards, and the artifacts that are inherent in the imaging system.

Study [95] presented the development of FaceQvec, a software component designed to estimate the conformity of facial images with the criteria outlined in ISO/IEC 19794-5. This international standard provides general quality guidelines for face images, determining their suitability for official documents like passports or ID cards.

5.4. Information Management Standard

The present information management standard delineates fundamental principles and obligations that must be adhered to in order to achieve efficient information management. The primary objective of this system is to ensure the proper creation, management, retention, and disposal of information.

5.4.1. Data Quality Standard

Establishing the principles of information and data quality and explaining the path to data quality can be used to manage the quality of digital data sets. This includes both structured data saved in databases and less structured data like images, audio, video, and electronic documents.

One article [96] estimates the ICT system IQ, and ISO information quality standards are examined. Due to the limits of existing IQ estimation approaches, the authors describe their own proprietary concept based on multidimensional and multi-layer modeling and uncertainty estimation.

Another article [97] proposes a custom user-centered web-based system based on lab requirements. By defining norms, standard processes, and procedures, supporting quality control and teamwork, and aiding project management, the implementation system improves efficiency and quality. A conclusion is reached that while the market for major manufacturers is a small niche, a tailored LIMS standard is crucial for heritage science lab administration worldwide.

5.4.2. Quality Management Standard

The following are essential elements that products, services, and processes must consistently adhere to in order to ensure that their quality aligns with expectations: requirements, specifications, guidelines, and characteristics. The items in question are suitable for their intended purpose. They effectively cater to the requirements of their users.

One paper [98] presented process harmonizations across remote sites as a “Best Practice” for similar organisation. Also, a process was defined to develop new tools and to select a standard set. CT DC realized common KPIs, training, customer management, and career landscape with shared roles and responsibilities.

Four papers [99–102] aimed to show an ontology called SRMO, which attempts to bring together all the terms and ideas related to RM and offer a more complete picture of risk. This paper offered three examples of how this theory can be used in the field of software engineering to show how useful it is. The SRMO was built based on analyzing, comparing,

and integrating various terms for RM used in industry-wide standards, frameworks, and models like ISO 31000, COBIT, PMBOK, and CMMI. The quality of the software is more important than anything else. Software quality traits show what software is all about. Proper management of the software creation process is a meaningful way to solve the software quality problem. Many standards for assurance of software quality management, like ISO 9001 and CMMI, have already been developed to help businesses set up quality management systems.

5.4.3. Structure of Management Standard

A management system is a collection of procedures that an organization must adhere to in order to achieve its objectives. A management system standard offers a structured framework for establishing and managing a management system.

One paper [103] assessed the maturity of agile adoption. The first step in adopting agile methodologies into automotive software product lines is to evaluate the combination. The ASPLA model was created to assess agile software development using ISO 26550 based on an interview study with 16 participants and a literature analysis.

5.4.4. Risk Assessment Techniques Standard

This standard offers assistance in the selection and implementation of systematic methodologies for the evaluation of potential risks. One study employs a variety of methodologies to identify and comprehend risk. The recent update has been implemented to broaden the scope of its applications and enhance the level of detail provided.

One paper [104] on a package is made better by the addition of a graphical tool for predicting the collection area. This tool offers the user a way to draw the structure, no matter how complicated it is. The software was implemented in the National Telecommunications Corporation building, which is the tallest building in Sudan. Compared to Furse's StrikeRisk v5.0, the package outperformed in calculating risk components and total risk.

5.5. Telecommunication Standardisation

Today's converged communications market is merging voice, data, video, and wireless technologies. As more firms join the dynamic communications network, they recognize that Standardisation, planning, and continuous development are essential for functionality, interoperability, and reliability.

The telecommunication Standardisation sector defines the basic transport and access technologies that power global communications networks. Standards power modern wireless, broadband, and multimedia technologies.

5.5.1. Wireless Networks Standard

Wireless standards encompass a collection of services and protocols that govern the behavior of Wi-Fi networks and other data transmission networks.

Paper [105] is concerned with terms of time and energy usage. This work re-examines the issue of tag identification in UHF RFID systems. Previous tag reading procedures are thoroughly evaluated and analyzed in this work. A new design of a tag reading algorithm based on prior art is discussed to improve the time and energy efficiency of the ISO/IEC 18000-7 EPC C1 Gen2 UHF RFID standard.

Three papers [106–108] make a lot of progress in using RFID technology in libraries over the last ten years. ISO 28560 is a new standard that aims to provide a shared platform for an industry that has grown up while different implementers have come up with their own ways to store RFID data. Most RFID standards, such as ISO 14223, 14443, and 15693, which deal with air interface protocols, were created before they were put into use. Standards like ISO 28560, however, which deal with standardizing tag data models, were created after the implementation reached a certain level of maturity.

IEEE 802.15.4 is another paper [109] on the study investigating the benefits of SDN in infrastructure-less wireless networking systems like W-PANs and the requirements for its use. SDWN, a complete SDN solution, is introduced and a prototype implementation described.

Two papers [110,111] on a framework for multipatient positioning in a wireless body area network (WBAN) utilize spatial sparsity and FFT-based feature extraction to track patients' movements and report to a central database server. This compressive sensing theory provides excellent precision, detail, and low computing complexity.

Two publications [112,113] expand band-wide per-subcarrier PMI selection. IEEE 802.16e-2005 link-level simulations are used to test the recommended methods. One precoder per subcarrier band minimizes overhead. This is an easy and effective way to make sure that extended-real-time polling service (ertPS) received the most out of its uplink resources in IEEE 802.16e broadband wireless networks. Based on the MAC request header numbers, the suggested algorithm dynamically offers uplink resources to subscriber stations whose base stations are receiving more requests.

Four papers [114–117] are concerned with a code stacking method that uses Walsh code to double the data rate. How well the system reduces inter-symbol interference (ISI) IEEE 802.15.6 determines how the amplitude-modulated FSDT technique is used. This study shows two ways to cut down on ISI. Increasing the data rate makes the signal-to-noise ratio 1 dB worse than with FSDT. The suggested interleaver design method doubles the amount of data that can be sent over 22 MIMO-OFDM communication systems that do not use transmit diversity.

There are three main contributions of this research: The novel proposals are introduced first. Second, the suggested IEEE 802.11p approach is used to calculate CTTs for each street, and SUMO is used as a traffic simulator for dynamic route planning [118].

Two papers [119,120] show that the idea that these techniques hurt performance is not true. Using the better SVD method to use channel prediction shows that it almost works as well as using a spatial filtering method with IEEE 802.11ac channel prediction. SVD is a good MIMO receiving method for channels that change over time.

In another study [121], the IEEE 802.15.4 standard focuses on understanding SDN benefits in wireless networking settings without infrastructure, like W-PANs, and requirements for SDN in W-PANs. SDWN, a full SDN solution, is explained, along with a prototype implementation.

Another study [122] proposes using AIC to reduce energy and enhance performance. With the Intel 82579 NIC and the e1000e Linux device driver, IEEE at the source with AIC is set up. The results of the experiments show that the energy efficiency of AIC is better in most cases, even when performance is taken into account. In the best case, it is up to 37% better than the energy efficiency of traditional interrupt coalescing methods.

In one paper [123], the statistical properties of 60 GHz channels in a staircase environment with the transmitter (Tx) side constant and the receiver (Rx) side moving are examined. Specifically, the paper looks at how the moving receiver Rx changes the properties. IEEE 802.15.3c shows a single-input-multiple-output (SIMO) model that can be used in an office or meeting room. Simulation results show that the Euclidean distance between the Tx and Rx is linked to almost every channel property.

One article [124] shows that holonic and CPS models work well together on the IEC 61499 standard for production control automation. This study uses UML models and a protocol based on industrial Ethernet (Profinet) to meet the needs of a distributed CPS that works in real time.

5.5.2. Networks Standard

A networking standard refers to a formally established document created to outline the necessary technical requirements, specifications, and guidelines that must be consistently adhered to in order to ensure that networking devices, equipment, and software are suitable for their intended purpose. Standards play a crucial role in guaranteeing the attainment of high levels of quality, safety, and efficiency.

There are four studies [125–128] that propose an approach to AP selection that actively accounts for both types of mobility. Furthermore, they demonstrate that the proposed method greatly outperforms the baseline method in terms of throughput. Because of the quick growth of WLANs based on the IEEE 802.11 standard, more and more public places like train stations and airports are offering WLAN services. This work introduces a new type of acknowledgement called TACK and the corresponding TCP extension, TCP-TACK. TACK's goal is to reduce the number of ACKs sent, which is exactly what the transport protocol needs. Without modifying the hardware of IEEE 802.11, TCP-TACK can be used on top of commodity WLAN to provide high wireless transport goodput with little control overhead in the form of ACKs.

Two publications [129,130] detail preliminary work on conforming network entities and progress toward a building-sized test bed for this specification. Rowan university students and researchers will have a living smart sensor network for development and study after initial work. A prototype system monitors LV networks near-real-time. It is explained how the first and second releases of this secondary substation automation equipment utilize IEC 61850 and CIM standards.

One paper [131] covers RFT-V and RFT-C specifications. It compares voltage, current, and power limits, as well as different deployment methods, cable pairs, distance served, etc. The publication concludes with safety measures for both standards, including UL/CSA/IEC6095021.

Another study [132] states that FRAND-committed SEP holders seek a higher royalty rate (holdup) and implementers want a lower rate (holdout). Court lawsuit 802.11 determines royalty rates if negotiations fail. SEP valuation must be calculated accurately. In this study, the top-down FRAND royalty rate is applied with the equivalent license strategy to supplement it.

NS2 implements another paper's [133] approach for wireless broadband network application. This new scheme evaluates ASVC on a more realistic simulation across wireless broadband networks using open-source software. MGS mode ASVC beats CGS mode IEEE 802.11n.

One paper [134] proposes ASBA to tackle IEEE 802.11 wireless mesh network congestion. The ASBA uses collision states to alter contention windows, improve wireless mesh network contention performance, and avoid delivery delays.

The performance of another paper [135] gateway is tried and looked at. Also, the program code is optimized to improve its performance ISO 11898-1 after a gateway study. After optimization, the findings show that the performance of the proposed gateway has improved significantly.

Four studies [136–139] propose a beam training algorithm with dynamic beam ordering that meets the strict latency criteria of the latest mmWave standard negotiations. Simulation results show that different variations of the proposed scheme improve latency performance and received signal-to-noise ratio over the optimum beam training scheme based on exhaustive narrow beam search IEEE 802.11ay. Driving data are obtained from V2X driving status information and predicted by LSTM. The proposed technique accurately estimates maneuver via V2X communications and follows proven V2X standards IEEE802.11p.

5.6. De-Facto Standard

De facto standards are those that originate in the marketplace and are endorsed by a number of suppliers but lack formal status. For example, Microsoft Windows is a single company's product that has not been formally recognized by any standards organization while being a de facto standard. De facto standards in the communications industry frequently become formal standards after widespread acceptance.

Even though the internet protocol is the de facto standard for internet connection, severely restricted devices with little processing power, working memory, and battery power capacity should be addressable via an IP-based network architecture. To meet this

requirement, 6LoWPAN could be used to communicate with wireless devices. Wireless sensor networks are a hot issue these days, but a well-structured, wire-based infrastructure is sometimes preferable [140,141].

According to one research [142], the technical advancement of wireless communication technologies, as well as the need for effectively modeling these increasingly complex systems, produces a continual increase in simulation model complexity. Simultaneously, multi-core systems have emerged as the de facto standard hardware platform.

Another study [143] discussed the HEVC standard which is now the de facto video coding standard. The OpenHEVC open source software is changed in this study to improve interoperability by using OpenMP instead of POSIX threads. This paper proposes a novel approach to dealing with parallelization throughout the decoding process. This method outperforms the POSIX-based one by at least 10% while providing greater interoperability.

Five publications [144–148] offer a programmable implementation and continuation of the same project in which behavior diagrams, namely use case, activity, and state diagrams, are chosen for critical study and possible enhancements. The unified modeling language (UML) has become the de facto standard for the design and specification of object-oriented systems, although its semi-formal structure has certain drawbacks.

Two papers [149,150] examine how a standards-based pro-competition legal interoperability framework can be used to keep future internet service marketplaces open, inventive, and competitive. The following is how the paper is organized: the specific issues generated by a shift in favor of dominant de facto standards in ICT markets are explained.

Another two publications [151,152] examine how FINEAS is a rate adaptation algorithm that enhances client QoE and fairness in multiclient contexts. It involves an in-network coordination system for equitable resource sharing. HTTP adaptive streaming (HAS) protocols have gained popularity and become the de facto standard for video streaming services.

Three articles [153–155] implement a subsumption architecture utilizing model-driven development on a real-time physical platform. The robot operating system, the current robotics standard, is compatible with the platform. This work experimentally implements a multi-layer subsumption-based autonomous robot control utilizing model-driven development. This paper proposes an alternate datatype generic algorithm for calculating the difference between any algebraic datatype's values.

Four articles [156–159] use runtime profiling and dynamic code creation to circumvent these query compiling restrictions. Apache Spark is becoming the norm for modern data analytics in big-data systems. Spark optimizes analytical workloads on various data sources through SQL query compilation. JavaScript is becoming the mainstream language for general-purpose programs after dominating client-side web programming. Most engineering fields model complicated systems.

Four studies [160–163] discuss a cross-platform, open-source, distributed version control tool that works well with non-linear development and can handle projects of all sizes quickly and effectively. It has become the de facto standard for version control in software creation and is used by millions of software developers every day. The decision to use Android SDK instead of platys is based on two things. First of all, the Android SDK is the usual way to build location-aware apps for Android.

One paper [164] is concerned with the fact that hardware layers, libraries, kernels, and light-weight operating systems are often used to help with development. But these usually do not help much or at all with automatic worst-case execution time (WCET) estimation, so manual methods based on testing and measuring remain the de facto standard.

6. Smart Technology Standard

The acronym SMART stands for the various formats, processes, and tools that are essential for facilitating user interaction with standards, encompassing both human- and technology-based users. In the nineteen articles in this category, these digital solutions cater

to the requirements of various stakeholders, including industry representatives, regulators, end users, and society as a whole.

6.1. Internet of Things (IoT) Standard

The IoT has challenges due to the lack of standards because it makes it challenging for various systems and devices to collaborate and communicate effectively. It is challenging to create one universal standard working across all IoT devices, as they might use a plethora of various protocols of communication and data formats and be manufactured by many different vendors.

Study [165] examines the signals obtained from discreet sensors integrated into a wheelchair which possess convenient plug-and-play functionality and automatic identification capabilities. These signals are collected and initially processed at the platform level and subsequently transmitted to a server application on a host PC using the IEEE802.15.4 wireless communication protocol.

According to study [166], enabling full connectivity among these devices is a huge task. To address this, various communication protocols have been proposed. IEC 61850 has become the de facto standard for sophisticated communication networks in power systems due to its capacity to handle enormous volumes of data exchanges and object-oriented design.

6.2. Smart Grids Standard

Smart grid technology and the products and services that use it offer a lot of potential for businesses in all fields. The number of successful smart grid projects keeps rising, and each one is different in terms of its size and scope. Even though there is not a single meaning of “smart grid” yet, we discuss the topic below.

Two articles [167,168] showed how the proposed modeling idea could be built on the power utility interoperability approach IEC 61850 and the distributed automation reference model IEC 61499. A lot of attention has also been given to the ICT standards, such as IEC 61850, which combines ICT with MAS.

Two other publications [169,170] presented a PnP approach to intelligent terminals in PDIIoT that takes network priority into account. To guarantee the integrity of PnP technology’s communications in the distribution network, this paper first implemented a network access strategy selection based on the priority function and SROS method. Next, an information model of the PDIIoT was built using the IEC 61850 communication standard, and the PnP information interaction mechanism was described in depth.

Paper [171] aimed to conduct an analysis on the alignment between digital twin architectures documented in existing literature and the reference architecture outlined in the ISO 23247 standard. This result was achieved through a mixed-methods research design setting: it considered the analysis of 29 digital twin architectures applied to manufacturing. These architectures were identified through a systematic review of literature comprising 140 peer-reviewed studies, a survey conducted with 33 participants, and four in-depth expert interviews carried out in a semi-structured format.

In one work [172], authors used the standard’s extended backus Naur form to create a model of the IEC 61131. Automated processes are needed for current industrial automation projects to facilitate faster upgrades, enhancements, and maintenance. Model-driven development is used in software engineering to reduce development time by maintaining consistent documentation, models, and code throughout the lifecycle of the project and by using code generators to rapidly produce large volumes of highly comparable code artifacts and code for a variety of target platforms.

Three more papers [173–175] showed how function block descriptions can be used to make very efficient code. The results of a benchmark set show that compared to other methods, our approach makes code that is much faster and smaller. IEC 61499 has been put forward as a standard for creating control and measurement systems for industrial processes. So, even though hard real-time performance is not guaranteed and the runtime

environment is run on top of a normal operating system, the performance is consistent and looks good for future implementations and extensions.

A different group [176–178] on the digital sample method was used to describe a reference system for calibrating electronic transformer testers. The program for calibration was made to meet the IEC 61850-9-2/LE standard. With a coverage factor of two, the electronic transformer testers' extended errors were estimated to be $5.4\text{E-}5$ for magnitude and 64rad for phase displacement.

6.3. Energy Storage Standard

The energy management standard has the potential to significantly improve global energy consumption. Offering management strategies to public and private organisation can improve energy efficiency, reduce costs, and increase overall energy performance. Effective energy management is the top concern due to its large energy savings and GHG emissions reduction.

One article [179] examined the idea of a holistic SoS element, with a focus on the ICT and automation principles used. It described the applicability of present electrotechnical and automation standards within the future critical infrastructure.

In another study [180], a model-driven implementation of a CEM that meets standards was explained. The proposed application method is better than other methods because it automatically works with standard data models and lets CEMS logic be expressed in a way that does not depend on the protocol. IEEE 2030.5 and OpenADR 2.0b are the standards that are taken into account in this application.

6.4. Artificial Intelligence Standard

Describing methodologies concerning assessing the efficiency of machine learning models, systems, and algorithms in relation to categorization tasks enables some of the important enablers for digital transformation. It covers AI, defines key terms, and provides a high-level strategy for deploying machine learning.

Standardisation of software and hardware platforms and the exploration of application scenarios are discussed to facilitate the industrialization and commercialization of edge AI systems. IEEE 3652.1-2020, the IEEE standard for federated learning (FL), was developed by the Learning Technology Standards Committee of the IEEE Computer Society. It offers guidelines for FL architectures and categories, considering data, user, and system perspectives, and identifies relevant application scenarios, performance evaluations, and regulatory requirements [181].

6.5. De Facto Standards

One document [182] on the Extensible Markup Language (XML) is recognized as the de facto standard for semistructured data representation and data exchange on the Web and in many other settings. Due to its flexibility and adaptability for data representation and communication across applications and services across different platforms, the XML has become the de facto standard for representing semistructured information.

Another paper [183] suggests using ROS as an alternative flexible framework for developing software for robots. It is an API, a library, and a rule collector initiated to simplify the difficult task of making complex, reliable behavior for robots working on a wide range of robotic systems. It has become the standard way of building applications for robots.

7. Security Techniques Standard

The fourth category included nineteen papers on a rising interest in developing techniques to facilitate the creation of privacy-conscious software systems standards in response to the escalating concerns surrounding privacy in software systems. Insufficient privacy measures incorporated into software system designs may lead to inadvertent disclosure of users' sensitive data, including personal health and financial information, thereby po-

tentially resulting in financial and reputational repercussions. Privacy engineering can be defined as a methodical endeavor to incorporate privacy considerations into the architectural design of an application.

7.1. Cybersecurity and Privacy Protection Standard

Standards are responsible for helping to protect against the growing problems of cyber risks and attacks, online scams, information and identity theft, and the breach of personally identifiable information. It offers groups ways to protect their sensitive and important information as well as personal information, no matter what business they are in or how they are set up.

One study [184] found that software developers' intention to follow PEMs is most strongly influenced by the PEM's usefulness to their work. These data illuminate software developers' attitudes toward PEMs. A growing field of research helps software developers build privacy into their systems. ISO/IEC PDTR 27550 is developing privacy engineering standards.

In another work [185], the enterprise accessibility compliance ontology added relevant elements to earlier ontologies. Many online sites offer assistance on accessibility for practitioners, but for a corporation trying to make products accessible in a continuously changing landscape of technologies and standards, accessibility is a complex problem EN 301 549 standard.

7.2. Information Security Management Systems Standard

The standard provides recommendations for enterprises of any size and from any industry to create, implement, maintain, and continuously improve an information security management system.

One paper [186] initially standardized ISO/IEC 18092 as a de facto standard for radio communication dominates this market and has amassed many applications, including use as a digital wallet and ticketing system, due to its convenience and versatility.

Another research [187] identified three public space robot properties. Crowds, social norms and proxemics, and misbehavior are these properties, as well as the ways in which these qualities affect robot safety. This article educated stakeholders on public robot safety. In particular, it contended that the ISO 13482:2014 standard is unsuitable for public area robot safety.

Five articles [188–192] covered data security standards and their impact on cloud computing and the Industry. ISO27001, SOA, BS7799, PRINCE2, PCIDSS, COSO, COBIT, OPM3, CMMI, and ITIL evaluate recent information about proactive damage prevention research and offer a GQM-based approach to detect and resolve information security control violations.

To solve this issue, an SDSec technique was described in work [193]. It is made up of a hybrid anomaly detection module and a multi-level security response module that work in tandem to secure the ICS field zones. According to the international standard IEC62443, the hybrid anomaly detection module inspects anomalous behaviors from the perspectives of network communications and physical process states.

Another work [194] on the possibilities of dealing with cross-cutting issues at the requirements level was concerned with aspektoriented requirements engineering. In this research, it was demonstrated how PETs can be expressed as early aspects and incorporated into a given requirements model using issue frames notation and industry standards such as ISO 29100. It was shown how PETs can be expressed as patterns to assist requirements engineers in identifying and selecting relevant PETs that address the privacy criteria they must meet.

One article [195] discussed how blockchain technology can be used to deploy building information modeling technologies. The possibility of implementing blockchain revealed that the technology is significant to an increasing number of actors. The paper presented

an investigation of the Polterovich–Tonis evolutionary model in connection to the ISO 16739: 2013 information modeling procedures.

In [196], the architecture was validated by building an industrial pick-and-place testbed using ISO/IEC 30141 to address interconnection issues and bridge the gap between the Internet and industry. A four-domain integrated architecture was proposed, dividing the manufacturing system into user, cloud service, sensing, controlling, and device domains.

Paper [197] used defeasible logic to establish compliance with software design and implementation constraints. Opportunities were reused from compliance checking. Finally, conclusions were drawn and future study was planned based on ISO 26262, which addresses cybersecurity as well as safety.

Another paper [198] described the primary safety communication protocol implementation aspects discovered during programmable logic controller development. Communication failures can endanger persons, property, and the environment; thus, vital industrial automation safety systems must detect as many issues as possible. Safety communication methods were designed and approved to IEC 61784-3 and 61508 decrease failures.

One paper [199] used ISO/IEC 9798 standards, key agreement, entity authentication, and hybrid algorithm technology to create a secure channel protocol for high-security data transfer and examine its complexity and performance. A security chip completed it. Data transmission tests revealed that it meets data communication security and real-time needs with its high security and flexible protocol.

7.3. De Facto Standards

One study [200] on differential privacy became the de facto standard for defining privacy. Companies like Apple, Google, Uber, and Microsoft use it to collect sensitive information about users and build analytics engines that protect privacy.

Another study [201] addressed how to accomplish low-cost security in real-world wireless sensor networks without changing security standards. Focused on ECDSA, the standard signature system is used in WSN and IoT applications to ensure communication authenticity and integrity.

Another study [202] on dynamic adaptive streaming over HTTP (DASH) is the standard for video streaming, and YouTube, Netflix, and other sites. In this paper, a way to identify videos by using network data while streaming is discussed.

8. Healthcare Informatics Standard

This category's twenty-five articles focus on the methodology for analyzing health organisation's business and information demands to support standards-based architectures, also highlighting the Standardisation of health ICT to allow for compatibility and interoperability amongst independent systems.

8.1. Medical Device Software Standard

The term medical device refers to any instrument, apparatus, implement, machine, appliance, or similar article, including software, that is intended by the manufacturer to be used for diagnostic and/or therapeutic purposes in human beings. These devices serve one or more specific purposes, such as diagnosis, prevention, monitoring, treatment, or alleviation of disease, as well as diagnosis, monitoring, treatment, alleviation, or compensation for an injury or disability.

One paper [203] examined the following topics: (1) the importance of EHRs; (2) the need for Standardisation to improve quality; (3) the importance of interoperability in EHR maintenance; (4) the open EHR architecture as a framework for Standardisation and interoperability; (5) the importance of archetypes for knowledge-based systems; and (6) the challenges of querying HER data in accordance with ISO 26262.

Two studies [204,205] proposed a medical-standard software development approach. This work discusses distributed simulation motivations, related studies, and IEC 82304's

justification. The standard's four types of interoperability reference models are offered. Healthcare and manufacturing examples show how the standard is used.

One paper [206] on the hybrid assessment approach discussed here combined the MDevSPICE®-based process assessment method with steps for prioritizing improvement needs through value stream mapping and allowing process improvement through the KATA technique. This method builds agile methods into the process of making software for medical devices while still meeting the requirements of the legal standards IEC 823041.

Another paper [207] on this topic asserted that ISO 14971 is the standard that regulatory agencies use to control the risks of medical devices. The ISO 14971 describes the steps of the risk management method and suggests that manufacturers use tools like the FMECA.

8.2. Device Interoperability Standard

Medical device interoperability refers to the capacity to exchange and utilize information among multiple devices, products, technologies, or systems in a safe, secure, and efficient manner. The shared information can be utilized in various manners, such as for display, storage, interpretation, analysis, and automated action or control of another product. With the growing integration of electronic medical devices with one another and other technologies, it is imperative that connected systems possess the capability to exchange and utilize information in a safe, secure, and efficient manner.

Three articles [208–210] proposed enhancing the ISO/IEEE 11073-20601 protocol with a biometric identity management system. The rise in healthcare prices over the previous decade has spurred the ICT industry to respond by developing cost-cutting solutions while enhancing healthcare quality. The recently established ISO/IEEE 11073 family of standards is the first step toward interoperability of mobile medical devices used in patient situations.

One study [211] described how technology might be utilized to tackle the problems of healthcare administration as outlined in ISO 18092. This technology is user-friendly and financially feasible, and it is also simple to use for nontechnical users. NFC cards, tags, and key chains are examples of NFC technology. Each NFC tag contains a unique number that can be used to identify patients in a hospital.

Six software [212–217] development papers explained these principles. After turning on the system, safety devices can accurately determine an axis position. The V-module of software development standards after ISO 26262 makes the sensor system nearly error-proof. ISO 26262 defines functional safety for all automobile electrical and electronic systems. Analyzing AUTOSAR application software component production code for an instrument panel cluster provides a comparison of top performance tools.

In four articles [218–221], fixed-priority scheduling of mixed criticality SR models was investigated utilizing adaptive and elastic MCS. The optimization problem of minimizing the overall system cost of increased functional delays in the implementation was formulated while ensuring schedulability in the automotive certification standard ISO 26262, and an optimal branch-and-bound search algorithm was provided together with an efficient heuristic approach. This technique minimizes project development time, standardizes, forecasts software problems, and enhances quality in OEM vehicle embedded software.

Five studies [222–226] discussed undergraduate software quality teaching. The National Institute of Technology Karnataka, Surathkal course covered aerospace standards, safety, software requirements, design, reviews, and testing. Teaching ARP 4761, ARP 4754, DO 178C, and DO 331 was insightful. ISO 26262 functional safety standard affects car design and assurance. ISO 26262 defines SEooCs. Finally, P-MCS was analyzed and shown for system schedulability, overheads, throughput, and predictability using ISO 26262 as a case study.

8.3. De Facto Standard

There is only one paper, [227], in this category. Since the current gold standard, NCBI BLAST, employs complicated heuristics, it is difficult to attain both great performance and perfect agreement. The results are shown for multiple use cases, and all agree 100% and

are faster than a fully parallel implementation of the reference code on a contemporary CPU by a factor of approximately $5\times$. Table 2 contains the summary of the review of the current works.

Table 2. Summary of the review of the current works.

No	Standard Topics	Sub-Topics	Findings	References
1	Review and survey articles	Systems and software engineering standard.	1- Taxonomy-based testing (TBT) standard is a novel testing approach for medical device software, utilizing three methodologies validated by industry experts. 2- Software engineering standards identified 12 factors and 52 sub-factors that could contribute to diverse situations among software development team members, highlighting potential contributors to diverse situations. 3- The standard is a foundation for organizational training and architecture management guidebooks. A preliminary model based on Agile governance best practices has been developed, providing partial results.	[12–20]
2	Review and survey articles	Smart technology standard	1- Engaged in discussions on interoperable IIoT standards, protocols, and models for digitizing the industrial revolution. 2- Blockchain technology standards in the energy sector is maturing and has potential for application across various sectors, facilitating the advancement of modern grid technologies.	[21–24]
3	Review and survey articles	Security techniques standard	1- Evaluates and analyzes secure software development methods and standards using surveys and comparisons. It aims to identify stages in the SDLC where these methods can enhance software application security, incorporating resilience, information quality, and vulnerability frameworks, ISO19650, and ISO27001. 2- The findings have led to the development of a design framework and a series of inquiries for designing, reporting, and evaluating effective user studies in end-user software engineering standards. 3- Techniques demonstrate compliance with software requirements for EN 50128 and IEC 60601-1 in the railway industry and medical field.	[25–38]
4	Computer-communication networks standard	Information technology fundamental standard	1- The software standards analyzes 3D-scanned electronic geometric models, providing specialized information equivalent to national regulatory texts, and discussing basic information completeness and presentation needs. 2- A two-level warning system, using visual and audible signals, effectively prompts drivers to reconsider their lane change decision, ensuring safety in accordance with ISO17387:2008 standards. 3- The analysis examines standard specifications and their implementation in open source software, focusing on two projects that utilize the PDF standard.	[39–49]
5	Computer-communication networks standard	Systems and software engineering standards	1- Introduced a modeling language using ISO/IEC 24744 metamodel, effectively representing textual information and demonstrating its potential in requirements engineering and various textual domains. 2- SystemC, an IEEE 1666 standard for embedded modeling, architectural exploration, performance analysis, software development, and functional verification, has been approved by the SA for RTL. 3- TooMVBC is a design framework that converts a verified computation model into executable MVBC code, ensuring compliance with IEC 61375 standards and addressing ambiguities during verification. 4- The serious game quality assessment method is an ISO/IEC 25040-aligned evaluation approach for serious gaming quality. It provides a usability and portability model matched to ISO/IEC 25010 standards.	[50–92]
6	Computer-Communication Networks standard	Information Representation Standard	1- A spatio-temporal model for archaeological data, based on ISO Standards 19107 and 19108, to ensure interoperability and use GeoUML for data analysis. 2- FaceQvec is a software component that assesses facial image conformity with ISO/IEC 19794-5 criteria, determining their suitability for official documents like passports and ID cards.	[41,93–95]
7	Computer-Communication Networks standard	Information management Standard	1- A spatio-temporal model for archaeological data, based on ISO Standards 19107 and 19108, to ensure interoperability and use GeoUML for data analysis. 2- FaceQvec is a software component that assesses facial image conformity with ISO/IEC 19794-5 criteria, determining their suitability for official documents like passports.	[41,93–95]

9. Discussion

This review presented pertinent studies on contemporary standards in the field. The primary objectives were to showcase a novel perspective on standards and emphasize current research trends within this domain. The survey findings unveiled three key elements pertaining to the literature’s content: the challenges encountered in effec-

tively utilizing these applications, recommended measures to mitigate these challenges, and the underlying motivations behind the proposed general framework for the search and browse procedure.

RQ3. What are the existing research gaps in the field of ICT Standardisation?

9.1. Challenges

According to the surveys, the researchers are interested in evaluating standards. There have been some significant research advances in the areas of study, assessment, contrast, and revision, but more improvements are needed in providing approaches across structures and systems, as well as employing an adequate, representative sample for surveys (see Figure 7).

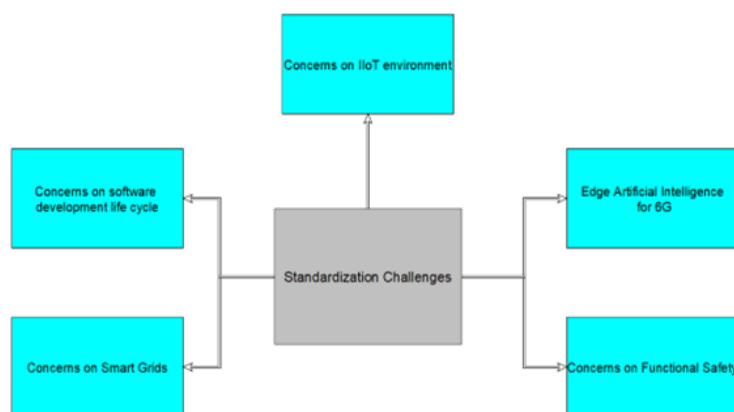


Figure 7. Categories of challenges.

9.1.1. Concerns on IIoT Environment

There are various hurdles to building an interoperable conceptual IIoT environment. (1) A huge number of heterogeneous IIoT devices share resources, and the bulk of IIoT devices are limited in terms of storage and computation. (2) In IIoT industrial applications and devices, there is a heavy dependency on a conceptual model that is composable and backed by a sophisticated orchestrator. Device tiering and cloud-fog technology orchestration are thus recommended for the IIoT ecosystem. Fog networking helps in industrial processing through quicker response to delay-critical industrial applications, while mature cloud technology can collect unlimited data and avails sharing resources that would be distributed across the IIoT environment. In time, the OpenFog Consortium was established to further refine and standardize the fog computing industry [21,228,229].

9.1.2. Concerns on Software Development Life Cycle (SDLC)

One of the most essential issues for businesses, and sometimes the most difficult one, is the development of safe software, as well as the implementation and management of security in the SDLC to make software more secure and adhere to standards and regulatory requirements to development standards. There is a set of software security standards, rules, and certifications to help guide secure software development projects. The survey to highlight their strengths and to provide an outline of where a standard may be most useful [230–232].

Faults and failure are two frequent ideas associated with any unsuccessful program [233]. Both defects and failure have a significant relationship. It is possible that software flaws to development standards remain unreported and do not result in a system breakdown. On the other hand, there may be some flaws that cause several failures. Faults in software engineering occur for a variety of causes, so there is a need for development standards. One of the causes of software engineering failures is a lack of understanding and carelessness in dealing with diverse situational settings among team members [234].

The importance of situational circumstances is recognized in software engineering standards. We explored the factors that can result in diverse situational scenarios, ultimately leading to software failure. Each element was subdivided into related sub-factors that can affect the effective development of a software system, resulting in software failure [235].

We drew the conclusion that numerous standards may not, taken separately, meet all the security needs for secure software development. Instead, more than one of the standards must be implemented as part of a software development process to ensure the security of the final product, whether for reasons of regulation or certification. Additional investigation into these facets, together with the solution ways for regulating them, can be carried out in subsequent work.

9.1.3. Concerns Regarding Smart Grids

The International Electrotechnical Commission (IEC) has introduced the IEC 61850 standard for power utility automation to ensure a significant level of interoperability, which is a crucial prerequisite for smart grids and their associated components and devices. A comprehensive approach to design, modeling, and implementation is necessary due to the limited scope of addressing interoperability and communication concerns [23].

The future grids exhibit a range of novel network and communication elements, such as inverters, smart meters, charging units, controller devices, switches, breakers, and others, which offer improved and advanced services. This results in the establishment of a highly intricate network comprising various interconnected and intelligent devices that necessitate the exchange of both information and energy. Intelligent Electronic Devices (IEDs) are highly sophisticated devices that provide advanced functionalities and services for effectively managing the intricate and decentralized characteristics of smart grids [236].

9.1.4. Concerns Regarding Functional Safety

Currently, there is a lack of a specific standard that directly addresses the functional safety requirements of autonomous vehicles. Consequently, numerous researchers have endeavored to employ the ISO 26262 standard as a framework for constructing software and hardware models that align autonomous vehicles with functional safety standards. The comprehensive examination presented various methodologies and approaches that can be employed to incorporate functional safety measures in autonomous and semi-autonomous vehicles [34].

Many approaches are still in development, indicating the necessity for a dedicated functional safety standard for autonomous vehicles. Moreover, the present ISO 26262 functional safety requirement for road vehicles is insufficient when considering the autonomous vehicle as more safety-critical and sophisticated. There is no need merely to update a new devoted standard for the functional safety of an autonomous vehicle, but the ethical concerns and risk-based fallacies of the present standard need to be addressed. New safety and security criteria for autonomous vehicles are urgently needed, and this is the scope of future research on this subject.

9.1.5. Edge Artificial Intelligence for 6G

The growth of services that use artificial intelligence (AI) is pushing wireless networks to change even more. There is a public notion that 6G will be a game-changer and will really fast-track the transition from “connected things” to “connected intelligence” in wireless communication around the year 2030. The article discussed Standardisation, software and hardware platforms, and possibilities for applications, making it much easier for the systems of mass-produced edge AI to be sold.

With the Standardisation and global rollout of 5G networks, researchers, businesses, and governments have started working on the vision, use cases, and disruptive technologies for future 6G. With the creation of wireless data collection, learning models and algorithms, as well as software and hardware platforms, it is possible that AI will become a natural

tool for designing disruptive wireless technologies to speed up the design, Standardisation, and commercialization of 6G [181].

9.2. Motivations

Standardisation allows for the production of differentiated products. Standardisation certainly aids businesses in expanding their networks. The Standardisation process generates a research output that is ultimately linked to the creation of markets for new technologies and can connect to a plethora of existing standards and technologies. Standards can help to establish a new technology and build its market since they are easily referenced (for example, in procurement decisions; see Figure 8).

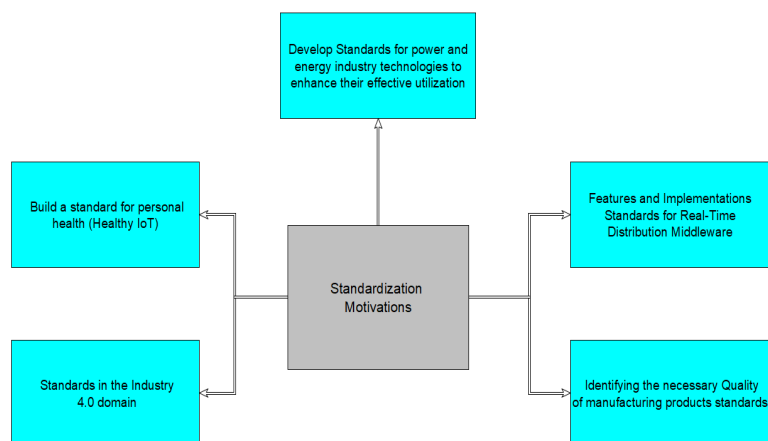


Figure 8. Categories of motivations.

RQ4. What are the key motivations, concerns, challenges, and recommendations for improving Standardisation and ICT methodologies?

9.2.1. Build a Standard for Personal Health (Healthy IoT)

Using the standards for personal health gadgets to build an end-to-end remote monitoring platform prompted one of the most recent innovations, the IoT, which aims to help with health and everyday life (Healthy IoT). Even though the bottom layers are coming together, the upper layers, data models, and names are still being worked out.

Ultra-low power wireless protocols, such as ZigBee, offer notable benefits in supporting IEEE 11073. Nevertheless, the ongoing advancement of technology inevitably exerts a significant impact on the prevailing standards. The recent release of Bluetooth 5.0 incorporates several notable enhancements, including routing capabilities, extended range, larger baseband frames, and improved frame delivery to dormant end devices [209,237].

Another invention based on 802.15.4 is a 6LoPan (low power IPv6 in resource restricted networks) profile. Adoption of IP-based transport layer technology might result in technology convergence, including Wi-Fi (802.11); interoperability would be based on bridging and routing IP networks; and administrators could be situated in the cloud. IEEE 11073 might be implemented directly on this transport layer.

Several organisations (Standardisation bodies, vehicle manufacturers, and academia) have recently collaborated to define standards for vehicular applications and build VANET-based communication systems. Dedicated Short-Range Communication (DSRC) is a wireless technology developed exclusively for usage in automobiles. These protocols (DSRC) enable high-speed data communication between two cars (V2V) or between a vehicle and a roadside unit (V2I or I2V) [118].

Comparing the results of traffic and VANET simulations, it can be shown that IEEE 802.11p is a very realistic standard. It is seen from the results that V2X communication can already be realized for the field with the deployment of the IEEE 802.11p standard in a very resource-saving way and increasing comfort on road networks.

Potential future research endeavors may involve the simulation of urban traffic patterns utilizing alternative Dynamic Route Planning (DRP) methodologies. Additionally, the utilization of network simulators and wireless technologies, such as Long-Term Evolution (LTE), can be explored in this context. Furthermore, future research endeavors can center on investigating the impact of the proposed methodology on diverse traffic patterns observed in major urban centers and countries, such as the United States and Japan, which are subject to distinct regulatory frameworks and policies that influence their traffic dynamics.

Electric vehicles (EVs) are being increasingly embraced and incorporated into intelligent transport systems (ITSs) in developed nations, driven by the ongoing pursuit of sustainable transportation alternatives that provide enhanced energy efficiency and diminished carbon emissions. The establishment of communication between electric vehicles (EVs) and roadside units (RSUs) in intelligent transportation systems (ITSs) and smart grids can be achieved through the utilization of onboard units designed for wireless transmission. This standardized integration framework establishes a strong foundation for the “plug and play” concept to diffuse through the Internet of Things domain in electric vehicles. The seamless integration of the technology and components between the smart grid and ITS—like CS and RSU—further supports the smart city concept [178].

The implementation of Intelligent Traffic Systems (ITSs) and Autonomous Driving (AD) technology has necessitated the development of the Cooperative Vehicle Infrastructure System (CVIS). This system mandates that all vehicles possess the ability to comprehend the immediate and surrounding traffic conditions. Specifically, this can help achieve actual, continual, and effective communication between Vehicle-to-Vehicle and Vehicle-to-Infrastructure networks. Coordinating the interaction of vehicles with road infrastructure can enhance traffic safety and driving services [120].

In addition, this coordinating technology with regard to roads requires reliable and efficient V2V/V2I communication technology so that it can be feasible to apply it in current traffic conditions. In the present context, it is anticipated that the Cardiovascular Information System (CVIS) will progress in the subsequent areas. The vehicle wireless communication network is characterized by its extensive scale and intricate nature. Consequently, the utilization of network simulators and co-simulation software becomes imperative in order to assess the network’s performance, validate algorithms, and establish standardized protocols. Undoubtedly, despite the considerable expense, it remains imperative.

In the European context, the primary focus of CVIS research revolves around leveraging advanced information and communication technologies to expedite the research, development, and integrated implementation of safety systems. This research places particular emphasis on investigating communication standards and standardizing interfaces.

9.2.2. Standards in the Industry 4.0 Domain

Industrial wireless standards and communication technologies are the focal point of this report. The analysis proceeds with a discussion of the wireless standard security mechanisms and a summary of their primary vulnerabilities and security issues. There is discussion on the presented topic, with recommendations and suggestions for implementing the various wireless standards in the domain of Industry 4.0. There are numerous wireless communication standards within the Industry 4.0 domain [25].

In certain ways, this allows for the selection of the best communication technology for each and every industrial application. The variability, however, poses a substantial security difficulty because no universal security methods can be applied to the networking layer of the communication hierarchy. Individual wireless communication technologies try to reduce this by implementing their own Physical, Link, and Network Layer security procedures; however, many remain vulnerable owing to design faults and weaknesses. All of this needs the creation of new unified security measures that are applicable to all wireless communication technologies in the Industry 4.0 arena.

9.2.3. Develop Standards for Power and Energy Industry Technologies to Enhance Their Effective Utilization

To address the existing gap, this report aims to offer a concise overview of grid and prosumer blockchain applications, as well as present the standard development activities associated with them. The analysis indicates that blockchain technology in the energy sector is evolving and has the potential to be applied across various domains within the power and energy industry. This has the capability to support the advancement of diverse contemporary grid technologies. The provision of standards and guidelines for blockchain applications in the energy sector will enhance the effective utilization of this technology [22].

The four most common blockchain network components are described below [238,239].

- **Consensus:** The consensus algorithm under consideration is a synchronization mechanism that instills trust within the blockchain network. The synchronization of ledgers is crucial as each node maintains its own copy. Consensus mechanisms are employed to guarantee uniformity of state across all distributed ledgers. The aforementioned mechanism has to possess qualities of reliability, functionality, security, and real-time capabilities. Proof of Work and Proof of Stake are two very well-known consensus protocols.
- **Peers:** “Peer” designates every node in the blockchain network. In such a way, through these peers, the ledger is maintained and updated, following verification of the transactions to be added to the ledger. The chief duties of a blockchain network peers are responsibilities for hosting both the ledger and execution of smart contracts.
- **Smart Contract:** A smart contract refers to a publicly accessible code that is mutually accepted by nodes and can be activated in response to specific events. Following the activation of the smart contract, predetermined transactions or processes are executed, thereby expanding the utilization of blockchain networks within business operations.
- **Ledger:** A ledger refers to a comprehensive record of transactions and processes within the blockchain network. In the past, ledgers were widely utilized for record-keeping purposes. However, in the context of a blockchain network, each node maintains its own copy of the ledger. These nodes update their copies with valid transactions and synchronize them using a consensus protocol.

9.2.4. Features and Implementation Standards for Real-Time Distribution Middleware

To identify a set of features and objectives that all real-time distribution standards and/or their implementations should include [12], the following should be considered.

- **Control in the Processing of Remote Calls or Data:** Irrespective of the concurrency pattern employed, ensuring the predictability of the application necessitates the regulation of scheduling parameters, including threads generated internally by middleware. Implementing this approach helps mitigate the risk of encountering unbounded priority inversions.
- **Allowing Free Assignment of Scheduling Parameters:** In order to maximize the system’s scheduleability, scheduling parameters should be freely assignable throughout the chain of entities that compose the distributed system.
- **Support for different scheduling policies:** While fixed-priority scheduling policy currently absorbs a considerable amount of popularity and wide acceptance, certain conditions might be found whereby either EDF or sometimes flexible scheduling based on contracts turn out to be more appropriate. Hence, real-time distribution standards should aim at offering uniform support for configuring different scheduling policies.
- **The middleware should provide sufficient support to guarantee the predictability of the distributed system, thus reducing the priority inversion effects.** One way is by limiting the number of network communications to a manageable level; another is by facilitating the setting of synchronization protocols for gaining access to crucial portions.
- **Documentation of implementation-induced overhead:** Practitioners should be able to assess and evaluate each entity participating in a distributed real-time application,

including those formed internally by middleware (e.g., internal threads for I/O management or network messages belonging to metatraffic). The implementation must clearly specify the role and influence of these built-in entities, as these entities can raise system response times by consuming processing and/or network resources. As a result, distribution standards should include it as a condition for implementation.

- Facilitating the analysis of timing for the entire application: The execution of middleware occurs within the processor; however, it is important to note that the temporal behavior of the networks significantly impacts the overall response times. Furthermore, it is often necessary to schedule both networks and processors simultaneously using suitable techniques. Consequently, middleware should possess the capability to define the scheduling parameters for both processing resources.

9.2.5. Identifying the Necessary Quality of Manufacturing Product Standards

In recent times, the significance of manufacturing product quality has emerged as a pivotal determinant impacting both domestic and global business and economic dynamics. Over the course of time, a multitude of quality standards have been formulated and embraced, among which the ISO series of standards stands as a global agreement on commendable managerial approaches. The primary objective of these standards is to guarantee that an organization is capable of providing products or services that align with the quality expectations of its clients. This advising system provides the business with assessment results and implementation recommendations by identifying the necessary ISO elements and comparing the company's present quality performance to ISO standards. These recommendations are made available in the form of corrective and preventative action reports [102].

These standards thus guide organisation in defining effective quality management and provide models against which the system can be audited. This auditing process ascertains the efficiency of the organization's operation and provides the required assurance to both the organization and its clients. This caachin be achieved by undertaking an exhaustive analysis of the key ISO components and performing a comparative analysis between a company's existing quality system and established ISO standards.

9.3. Recommendations

Standards have the potential to meet most of the requirements in many different applications. Standardisation frameworks provide a common language for describing specifications of products, services, and processes. Section 6 presents a set of guidelines to address several issues and challenges associated with Standardisation (see Figure 9).

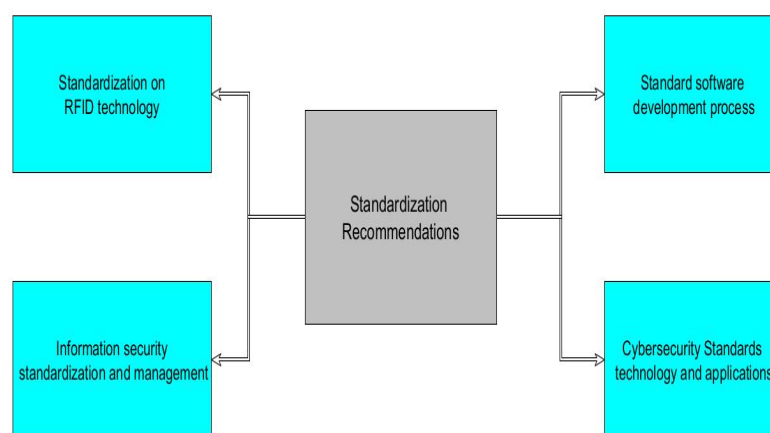


Figure 9. Categories of recommendations.

RQ5. How can standards play a more prominent role in achieving regulatory objectives and fostering innovation?

The standards are aligned to regulatory goals; this is because they bring in with themselves consistency, clarity, efficiency, innovation, and market access. This makes for a well-balanced environment between compliance and innovation.

9.3.1. Implementation of Standardisation RFID Technology

The implementation of RFID technology has made significant advancements in the past decade. The implementation of Standardisation within the RFID market has proven to be advantageous for the industry. The RFID suppliers are therefore able to sell their innovative products without significant entry barriers from competitors. Libraries have also benefitted by not being tied to any one supplier, and thus being able to select and combine solutions that best meet their needs [107].

The utilization of Radio Frequency Identification (RFID) technology spans across various industries, thereby necessitating intricate standards in this domain. From a categorical perspective, RFID standards can be classified into the following categories: technical standards and data content and coding standards. We adhere to a performance and consistency standard and strongly encourage the use of standardized practices. It includes the standard electrical characteristics associated with RFID technology, ranging from frequency and data formats to communication protocol, safety, testing, and application [240,241].

The testing standards for information technology play a crucial role in ensuring the quality and performance of equipment. The ISO/IEC RFID standard system incorporates test methods for equipment performance and conformance. These standards include but are not limited to testing competence as stipulated in ISO/IEC 18000, radio frequency identification protocol, among others [242].

9.3.2. Information Security Standardisation and Management

Ensuring information security is crucial for organisation and companies of all sizes, as it directly impacts the confidentiality, integrity, and availability of their valuable information assets. The absence of information renders operations unfeasible, while the presence of accurate and timely information enables the execution of various tasks and facilitates informed decision-making. Anticipating future advancements in the field of information security is a challenging task. However, it is evident that there is a pressing need for ongoing enhancements in the Standardisation and management of information security. The rationale for this is evident: the implementation of standards and good practices is necessary not only for internal efficiency and partner confidence, but also due to regulatory requirements. Media reports have brought to light numerous instances of information security management negligence [189].

9.3.3. Standard Software Development Process

Process adaptation may be necessary to both implement a standardized approach within an organization or project and to adjust a process to accommodate changing situational contexts, which are considered to be inevitable. Therefore, it would be advantageous for the software engineering community to have access to a comprehensive knowledge base that can aid in our comprehension of the mapping between software processes and specific situations. The creation of such a resource necessitates significant contributions from experts in software process. Additionally, it would greatly benefit from the inclusion of field data obtained from companies pertaining to both phenomena [64].

Software professionals can utilize this resource to enhance their software process adaptive capability in order to effectively address changing situations, resulting in improved process optimization. However, it is important to note that there is a significant level of complexity involved in the interaction between situational contexts and software processes. Therefore, any efforts to understand and evaluate this relationship should be approached with caution and meticulous attention to detail while also establishing a clearly defined

scope. The subsequent incremental action could entail the identification of the more critical components of the software process and situational factors, with an attempt to model their interaction.

Future research could focus on improving understanding of the relationship between specific situational conditions and features of the software development process. It has been observed that typical software processes must be modified to be acceptable for specific contexts [243]. Process tailoring is a word used to describe the practice of “adjusting the definitions and/or particularizing the terms of general description to derive a description applicable to an alternate (less general) environment” [244].

It has been argued that the wide range of IT projects presents challenges in attempting to establish a standardized approach for their development processes. The notion of a singular size that can accommodate all individuals is inadequate. Frequently, departure from a conventional methodology is perceived as a flaw, as an undesirable concession (despite its inevitable occurrence) [245].

9.3.4. Continuous Improvement of Systems in Maintenance Using a Proactive Quality Management

It is imperative to acknowledge that the proposed processes, as outlined in the standard, encompass all activities ranging from software specification to discontinuation. It is worth noting that maintenance is carried out promptly upon receipt of requests, without any provision for proactive measures to identify problems based on received feedback.

In light of this gap, a process to assist businesses who work with softwares in the maintenance phase, with a different approach than traditional ones, acting proactively in problem management, thus treating feedbacks and software user perception, has been developed. The goal is to provide answers to queries such as the following: “Does the software meet the needs of end users?”; “Where can software and the development process be improved to maintain product quality?”

9.3.5. Cybersecurity Standards Technology and Applications

Cybersecurity standards in the context of the operating system are the following: practical aspects, analysis, and comparisons.

The process of choosing security standards and deriving requirements from those standards within an organizational setting can be a laborious undertaking. Cybersecurity refers to the technological measures implemented to protect digital assets from unauthorized access and exploitation by malicious entities. The scope of these malevolent entities encompasses both individuals with authorized access to information systems and well-organized hacker collectives who possess significant motivation and resources, utilizing the internet as a medium to perpetrate attacks on said systems. The domain of cybersecurity is of growing importance due to our escalating reliance on computational devices [192,246].

The application of multiple standards is limited due to the possibility that a single standard may not adequately meet the security requirements of a specific organization. Consequently, additional efforts are necessary to integrate and align multiple standards in order to address the security needs of these organisation.

10. Conclusions

This paper reviews recent research on Standardisation and ICT, especially concerning education research. This paper provides an overview of standards in peer-reviewed publications, covering concepts like Standardisation bodies, standards catalogue, and hardware and software compatibility. It also discusses the increasing focus on software standards to address privacy concerns. The paper also discusses the role of health ICT Standardisation in interoperability and compatibility across diverse systems. The findings highlight the obstacles to effectively using these applications, recommended measures to address these challenges, and the motivations behind a general framework for the search and browsing

process. Another important benefit pointed out by this paper is that Standardisation helps in planning the whole research and innovation process by promoting discussions related to the exact outputs that the research is expected to deliver. The study also demonstrates that references pertaining to Standardisation within the call topics serve as a significant motivating factor in the decision to incorporate Standardisation. The review presents a comprehensive examination of the benefits of standard implementation, showcasing how these standards enhance various aspects of the research and innovation process. Key contributions include:

- A review of the benefits of standard implementation: the paper offers a detailed exploration of how standards enhance various research and innovation processes across multiple domains, highlighting the importance of their integration.
- Improvement of the assessment of research proposals: the evaluation process, upon incorporating set standards, strengthens the basis on which research proposals are evaluated to ensure that they meet international standards and are ready for implementation.
- Evaluation of existing standards: the review systematically analyzes current standards, providing critical insights into their effectiveness and areas for improvement.
- Fostering a global dialogue regarding the results of research: Standardisation helps improve the standards and update them to match technological advancement. Standardisation achieves efficiency in the distribution of research output by allowing for coverage of a wider audience while maintaining consistency in communicating findings.
- Effective dissemination of research findings: standards facilitate collaboration globally, hence narrowing gaps for effective international discourse. Standards establish a framework that facilitates interaction among scholars, institutions, and business.
- Encouragement of the implementation of innovative solutions: the implementation of standards guarantees the broader use of an invention while maintaining adherence to quality and safety norms, hence expanding its applicability across diverse markets.
- Guarantee of market compatibility and interoperability: the study points out how standards contribute to compatibility and interoperability of the products or solutions in the market, thereby reducing the barriers to entry, hence allowing seamless integration.
- Facilitation of the widespread dissemination of research findings: standards can also help raise the overall impact of the research by offering guidelines for timely and wide dissemination of the research findings to various stakeholders.
- Employment of a robust system to systematically document knowledge: an adequate documentation system of knowledge with preestablished methods guarantees that the result of research is kept, arranged, and accessible for subsequent continuous improvement and development during research.

Some individuals may lack a comprehensive understanding of the regulations set forth by Standardisation bodies, particularly in relation to accessing standards and the associated fees for participating in technical committees. It is believed that these concerns can be resolved by enhancing the governance and execution of Standardisation procedures. The issue of timing was brought up, as it has been observed that in the majority of projects, Standardisation efforts are initiated once significant progress is made, rather than being incorporated into the project from the outset. While acknowledging its significance, the subject of ICT Standardisation and its methodologies continues to be challenging to comprehend for many individuals. It appears that this field is becoming progressively exclusive to professionals and continues to be enigmatic to those without expertise. Research is published in this field; there is a need for general textbooks that can put the subject in a readable format for students of complete comprehension. We support increased emphasis on standards, particularly in the field of ICT. The urge for integrating ICT standard into classroom teaching will effectively empower the student with knowledge that addresses the challenges of this very significant industry. More importantly, inculcating elements of Standardisation within an education system means equipping the students with better abilities to handle the complexities arising in modern technological ecosystems. It might also enable their contribution to advancements within this area.

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Abbreviations

The following abbreviations are used in this manuscript:

IIOT	Industrial Internet of Things
IEEE(SA)	IEEE Standards Association
SAE	Society of Automotive Engineers
ARINC	Aeronautical Radio, Incorporated
IVHM	Integrated Vehicle Health Management
PHM	Prognostics and Health Management
DAECO	Digitised Architecture, Engineering, Construction and Operations
CPM/PDD	Critical Path Method/ Property-Driven Development/Design
gPROFIT	Technological Tool Based on a Model-Driven Theoretical Proposal
CMMI	Capability Maturity Model Integration
MAS	Multi-Agent Systems
PnP	Plug-and-Produce
PDIoT	Principles and Design of IoT
SROS	Self-Regulatory Organization
PEM	Privacy Engineering Methodologies
WLAN	Wireless Local-Area Network
LV	Low Voltage
SoS	System-of-a-System
CEM	Standard-Compliant Customer Energy Manager
LIMS	Laboratory Information Management System
EHRs	Electronic Health Records
NFC	Near-Field Communication
PETs	Privacy Enhancing Technologies
SDSec	Software-Defined Security
GQM	Goal, Question, Metrics
RTL	Register Transfer Level
TDL	Test Definition Language
CERTICSys	Software Assessment Methodology
SIMDECS	Medical Training Simulator
VSEs	Very Small Entities
ACO	Ant Colony Optimization
RPG	High-Level Programming Language for Business Applications

MCS	Mixed-Criticality Scheduling
SEooC	Safety Element out of Context
FMECA	Failure Mode Effects and Criticality Analysis
SRMO	Software Risk Management Ontology
COBIT	Control Objectives for Information and Related Technologies
PMBOK	Project Management Body of Knowledge
FPGA	Field Programmable Gate
OFDM	Orthogonal Frequency-Division Multiplexing
GTEM	Gigahertz Transverse Electromagnetic Cell
NCA	Neighborhood Component Analysis
CST	Chi-squared Tests
MRMR	Minimum Redundancy Maximum Relevance
P-MCS	Practical-Mixed-Criticality Systems
ASVC	Adaptive Scalable Video Coding
MGS	Medium Grain Scalable
CGS	Coarse Grain Scalable
ASBA	Adaptive State Backoff Algorithm
RFID	Radio Frequency Identification

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